



COMSATS University, Islamabad Pakistan

MediFusion Vision

By

AIZA KHADIM FA22-BCS-010

AREEBA NIAZI FA22-BCS-014

AREEHA NAYAB FA22-BCS-015

Supervisor

Dr. Rasool Bukhsh

Bachelor of Science in Computer Science (2022-2026)

The candidate confirms that the work submitted is their own and appropriate credit has been given where reference has been made to the work of others.



COMSATS University, Islamabad Pakistan

MediFusion Vision

**A project presented to
COMSATS University, Islamabad**

**In partial fulfillment
of the requirement for the degree of**

Bachelors of Science in Computer Science (2022-2026)

By

Aiza Khadim	CIIT/FA22-BCS-010/ISB
Areeba Niazi	CIIT/FA22-BCS-014/ISB
Areeha Nayab	CIIT/FA22-BCS-015/ISB

EXECUTIVE SUMMARY

In today's healthcare environment, there is a growing need for accessible, timely, and accurate diagnostics and remote consultations, especially in public clinics, hospitals, and underserved areas. Traditional methods of consulting doctors, reviewing patient reports, and managing medical appointments are often fragmented, slow, and depend heavily on manual processes. These inefficiencies may lead to delayed diagnosis, limited patient-doctor interaction, data silos, and increased workload for healthcare providers.

To address these challenges, MediFusion Vision is developed. MediFusion Vision is an integrated, AI-powered medical platform that combines patient, doctor, and admin interfaces for seamless healthcare management. The system enables patients to upload medical scans, book virtual or in-person appointments, and receive AI-assisted diagnostic reports. Doctors can access consolidated patient data, leverage AI insights for diagnosis, and conduct telemedicine consultations—all from a unified dashboard.

MediFusion Vision utilizes advanced AI models to analyze medical images, generate diagnostic findings, and highlight risk assessments, all presented in easy-to-understand reports for both clinicians and patients. The web and mobile applications provide role-based access and tailored experiences for patients, doctors, and healthcare administrators, streamlining the end-to-end process from diagnosis to follow-up.

The platform is built using modular web and mobile technology, secure cloud infrastructure, and adheres to healthcare data privacy standards. AI model monitoring, appointment logs, feedback review, and notification management are embedded for reliability and transparency. Patients benefit from reduced wait times, reliable records, and continuous access to expert advice, while doctors and admins save time on repetitive tasks and gain actionable insights from centralized analytics.

MediFusion Vision aims to bridge the gap between automated diagnostics and human expertise, bringing affordable, high-quality healthcare to diverse populations, and shaping a future where accurate, timely medical care is accessible to all.

Abstract

The proposal aims to create an intelligent system that assists doctors in detecting and diagnosing multiple critical diseases such as Brain Tumors, Alzheimer's, Stroke, Diabetic Retinopathy, AMD and Glaucoma through AI-powered analysis of medical images and patient data. The proposed system will analyse brain Magnetic Resonance Image MRI scans, retinal images and textual medical reports using advanced AI for efficient and accurate diagnosis of these diseases and will generate detailed reports. Both doctors and patients will be able to access these reports and additional features such as follow-up scheduling, doctor's feedback, and the patient's medical data. This platform will act as a comprehensive management system, organizing patient data and enabling easy access for both users. As far as we know, no system currently exists that can detect multiple diseases from one scan, while also managing patients and doctors in one place. This project aims to provide an efficient, accessible, and comprehensive solution that will improve the overall healthcare experience for users nationwide

Table of Contents

Abstract	3
1 Chapter 1: Introduction and Problem Definition	6
1.1 Overview of the Project	6
1.2 Vision Statement	6
1.3 Problem Statement	7
1.4 Problem Solution	7
1.5 Objectives of the Proposed System	8
1.6 Scope	8
1.6.1 Limitations/Constraints	9
1.7 Modules	9
1.7.1 Module 1: User, Role & Admin Management	9
1.7.2 Module 2: Patient Medical History & Consultation Notes	9
1.7.3 Module 3: Appointment & Schedule Management	9
1.7.4 Module 4: Diagnostic & Scan Management	10
1.7.5 Module 5: AI Diagnostic Engine	10
1.7.6 Module 6: Explainable AI & Transparency	10
1.7.7 Module 7: Final Report & Summary Generation	10
1.7.8 Module 8: Virtual Consultation Management	10
1.7.9 Module 9: Payment Gateway Integration	10
1.7.10 Module 10: Multi-language Support	11
1.7.11 Module 11: Feedback & Continuous Learning Module	11
1.7.12 Module 12: Accessibility & Assistive Tools	11
1.8 Related System Analysis/Literature Review	11
1.7.1 Literature Review	11
1.7.2 Related System Analysis	12
1.9 Tools and Technologies	13
1.10 Project Contribution	14
1.11 Relevance to Course Modules	15
2 Chapter 2: Requirement Analysis	16
2.1 User Classes and Characteristics	16
2.1.1 Use case Diagram	17
2.1.2 Requirement Identifying Technique	21
2.1.3 Backend Process Requirements (Event–Response Table Technique)	22
2.2 Functional Requirements	24
2.2.1 Recommended Format:	24
2.3 Non-Functional Requirements	65
2.3.1 Reliability	65
2.3.2 Usability	66
2.3.3 Performance	66
2.3.4 Security	66
2.4 External Interface Requirements	66
2.4.1 User Interfaces Requirements	67
2.4.2 Software interfaces	67
2.4.3 Hardware interfaces	69
2.4.4 Communications interfaces	69

List of Figures

Figure 2-1: Use Case Diagram for Admin.....	18
Figure 2-2: Use Case Diagram for Doctor.....	19
Figure 2-3: Use Case Diagram for Patient.....	20
Figure 2-4: Login Page	24
Table 2-3: Functional Requirements derived from Mockup M1	25
Figure 2-6: Dashboard Screen – Doctor Web.....	26
Figure 2-7: AI Diagnostic Workflow Screen.....	27
Figure 2-8: Appointment Management Screen.....	28
Figure 2-9: AI Note Taker Screen	29
Figure 2-10: Virtual Consultation Screen	30
Figure 2-11: Patient Details Screen	31
Figure 2-12: Patient List Screen	32
Figure 2-13: Final Diagnostic Report Screen	34
Figure 2-14: Patient Registration Screen	35
Figure 2-15: Patient Dashboard Screen	36
Figure 2-16: Doctor Search Screen.....	37
Figure 2-17: Doctor Profile Screen.....	38
Figure 2-18: Appointment Booking Screen.....	39
Figure 2-19: Payment Confirmation Screen	40
Figure 2-20: Scan Upload Screen	41
Figure 2-21: AI Analysis Results Screen.....	42
Figure 2-22: Feedback Screen	43
Figure 2-23: Dashboard Overview – Admin	44
Figure 2-24: Doctor Management Screen.....	45
Figure 2-25: Patient Management Screen.....	46
Figure 2-26: Appointment Logs Screen.....	48
Figure 2-27: Scan Repository Screen	49
Figure 2-28: AI Model Monitoring Screen.....	50
Figure 2-29: Notification Management Screen.....	51
Figure 2-30: Payment & Transaction History Screen.....	52
Figure 2-31: Feedback Review Screen	53
Figure 2-32: System Settings Screen	54
Figure2-33: Splash Screen – Mobile.....	55
Figure 2-34: Authentication Screen – Mobile	56
Figure 2-35: Home Dashboard Screen – Mobile	57
Figure 2-36: Find a Doctor Screen – Mobile	58
Figure 2-37: My Appointments Screen – Mobile	60
Figure 2-38: Book Appointment Screen – Mobile	60
Figure 2-39: Upload Scan Screen – Mobile.....	61
Figure 2-40: AI Report Screen – Mobile	62
Figure 2-41: Notifications & Profile Settings – Mobile	64
Figure 2-42: Consultation Chat Screen.....	64

1 Chapter 1: Introduction and Problem Definition

This chapter provides an overview of the MediFusion Vision project, describing its purpose, background, and the healthcare challenges it seeks to address. It introduces the motivation behind developing an AI-powered medical diagnostic system that combines image analysis and report interpretation to assist doctors in making faster and more accurate decisions. The chapter discusses the growing need for intelligent diagnostic tools in hospitals due to the increasing number of patients, shortage of specialists, and limitations of manual diagnosis. It highlights the shortcomings of existing systems that often lack automation, transparency, and integration between multiple disease detection processes. Furthermore, it defines the main problem, the targeted users — including doctors, patients, and hospital administrators — and the core objectives that guide the development of MediFusion Vision. Overall, this chapter lays the foundation for understanding the system’s importance, innovation, and the intelligent approach used in its design and implementation.

1.1 Overview of the Project

MediFusion Vision is an intelligent, AI-driven medical diagnostic platform designed to assist doctors and patients in improving diagnostic accuracy, efficiency, and collaboration through multimodal data analysis, explainable AI, and automated clinical documentation. The system falls under the domain of healthcare technology and artificial intelligence in medical diagnostics and is targeted toward healthcare professionals, patients, and clinical administrators seeking faster and more transparent diagnostic processes.

The proposed system integrates AI-based image and report analysis, virtual and physical consultations, and real-time speech-to-text transcription to enhance medical decision-making and reduce manual workloads. It leverages advanced deep learning models to analyze brain MRI and retinal scans for conditions such as Brain Tumor, Stroke, Alzheimer’s, Diabetic Retinopathy, Glaucoma, and Age-Related Macular Degeneration (AMD). Through explainable AI tools like Grad-CAM and SHAP visualizations, doctors can interpret the reasoning behind AI predictions, fostering trust and accountability in diagnosis.

In addition, the platform allows patients to book appointments, upload scans, view diagnostic reports, and communicate directly with doctors through secure virtual consultations. Administrators oversee the system by managing user registrations, linking hospital scans, and monitoring overall performance. By unifying diagnostic intelligence, patient-doctor interaction, and administrative oversight into a single platform, MediFusion Vision enhances healthcare accessibility, accuracy, and transparency, setting a foundation for next-generation AI-assisted medical systems.

1.2 Vision Statement

For doctors and patients **who** need a faster and more reliable way to detect and manage neurological and retinal diseases, **Medi Fusion Vision** is a web and mobile-enabled system **that**

assists doctors in making accurate, timely diagnoses while also providing patients with an easy-to-use interface for booking appointments, viewing medical reports, and maintaining their health records in one place. **Unlike** traditional diagnostic practices that often involve delays and heavy reliance on manual effort, **our** system uses advanced AI models to automate disease detection, reduce diagnostic errors, and improve overall healthcare efficiency. By minimizing the workload on doctors and accelerating the diagnostic process, the tool aims to increase survival rates through early detection and better patient engagement.

1.3 Problem Statement

In the medical field, one of the most overlooked issues is the occurrence of diagnostic errors. According to a study by Johns Hopkins University, nearly every individual is likely to experience at least one significant diagnostic error in their lifetime. It is estimated that 800,000 people in the United States alone either dying or becoming seriously disabled each year due to such errors. Neurological and vision-related disorders are particularly impacted by delayed or incorrect diagnoses. For example, Multiple Sclerosis (MS) affects over 2.8 million people worldwide, where late diagnosis can lead to irreversible nerve damage. Similarly, Alzheimer's Disease, a Neurogoical disorder, currently impacts more than 55 million people globally. Hence, early diagnosis is essential for effective intervention and care planning. Another similar disease 'Stroke' remains the second leading cause of death worldwide and the third leading cause of death and disability combined, accounting for approximately 11% of total global deaths. Prompt detection plays a vital role in improving patient recovery and survival outcomes.

Similarly retinal diseases major cause of low vision or blindness which effects huge population globally. These also can be cured if detected early. According to the World Health Organization, diabetic retinopathy is a major cause of adult-onset blindness, significantly affecting the diabetic population. Glaucoma, another widespread condition, impacts over 80 million individuals globally and can result in permanent vision loss if not detected in time. Furthermore, age-related macular degeneration (AMD) is expected to affect nearly 196 million people by 2020, with projections indicating further growth. It highlights the importance of early diagnosis for vision preservation. Despite the urgency, healthcare professionals often face challenges in diagnosing these conditions accurately due to rising workloads, limited time, and the complexity of interpreting diverse medical data. In the era of AI, there is an increasing demand for an intelligent diagnostic support system that can assist doctors by detecting these diseases early—before they lead to severe or irreversible harm

1.4 Problem Solution

The proposed system provides an AI-backed medical diagnostic solution that addresses the challenging task of detecting and diagnosing neurological and retinal diseases in their early stages. The system design is developed in a way that it assists doctors in analyzing retinal images, MRI scans and clinical reports which not only allows accurate diagnosis but also provide descriptive diagnostic reports. The application helps doctors detect serious diseases like brain tumour, Alzheimer's disease, stroke, diabetic retinopathy, glaucoma, and AMD more quickly and accurately, so that treatment can start on time and serious damage can be prevented.

1.5 Objectives of the Proposed System

- **BO-1:** Facilitate disease detection in early stages and offer descriptive diagnostic reports, improving overall treatment outcomes and patient survival rates.
- **BO-2:** Reduce the diagnostic burden on healthcare professionals by providing intelligent support for faster and more accurate decision-making.
- **BO-3:** Reduce the cost, time and human error for diagnosis of critical diseases.

1.6 Scope

Our project aims to develop both a web-based and mobile-based medical diagnostic application to detect and diagnose acute diseases like brain tumors, Alzheimer, Diabetic Retinopathy, Glaucoma and Age-related Macular Degeneration. It will also support diagnosis through textual medical reports.

The scope includes core functionalities for doctors, patients and admin users. The Admin will handle system setup and user management, including registering doctors and patients and managing internal scan uploads. Doctors will register based on their specialization (e.g., neurologist, ophthalmologist) and will have personalized dashboards showing appointments, patient profiles and their diagnostic reports.

Patients will register independently, create a basic profile, and book either physical or virtual appointments with doctors based on specialization, experience, and availability. The system will support both internal scan uploads by the admin and external uploads by patients via the portal. All relevant data, including appointments, reports and scans, will be securely stored and linked to the patient's profile in the database.

During consultations, doctors will record the patient's medical history if required, add notes, and request further diagnostic scans or tests. On follow-up visits, doctors can review the scans and run integrated AI models that analyze medical images or textual reports. These models detect and diagnose multiple diseases at once. With the help of Explainable AI (like GradCAM), they also show highlighted areas in medical scans that influenced the result. This helps doctors understand the AI's reasoning, validate it, and finalize the diagnosis with confidence.

The system then generates a comprehensive diagnostic report, combining AI output and the doctor's expert interpretation. Both are saved in the database and made available to patients through their portal.

The application enforces role-based access control to ensure that sensitive information is accessed only by authorized users. Doctors have the authority to review and update diagnostic outcomes, while patients have controlled access to reports, summaries, and doctor notes. Additionally, patients can initiate follow-up appointments and view their diagnostic reports.

1.6.1 Limitations/Constraints

- **LI-1: Image and Data Quality Dependency**

The diagnostic accuracy of MediFusion Vision heavily relies on the clarity and resolution of uploaded medical scans or reports. Poor-quality MRI or retinal images may reduce AI performance and lead to less reliable diagnostic outcomes.

- **LI-2: Connectivity and System Performance**

Since the system depends on cloud-based AI inference, real-time consultations, and data synchronization, a stable internet connection is essential. Network interruptions or limited bandwidth may impact the responsiveness of virtual consultations and delay diagnostic report generation.

- **LI-3: Dataset and Model Generalization Constraints**

AI models are trained on publicly available and institutionally approved datasets. While they perform well across general medical cases, variations in demographic or regional data may affect model generalization and diagnostic precision for underrepresented populations.

1.7 Modules

1.7.1 Module 1: User, Role & Admin Management

FE-1.1: Patient Registration & Login

FE-1.2: Doctor Registration & Login (with specialization & experience; approval by Admin)

FE-1.3: Admin Login and full access to user and system controls

FE-1.4: Role-Based Dashboard Access (Patient, Doctor, Admin)

FE-1.5: Admin can approve, reject, or manage registered doctors

FE-1.6: Patients can view/search doctors by specialization, experience, and availability

FE-1.7: Secure logout, session management, and profile editing

FE-1.8: Admin can review patient profiles, upload internal test reports, and monitor activity logs

FE-1.9: Admin can perform database backups and audit trails

1.7.2 Module 2: Patient Medical History & Consultation Notes

FE-2.1: AI note-taker record doctor-patient conversation during appointments, including symptoms and history.

FE-2.2: AI generates consultation notes and observations, which doctors can review and edit.

FE-2.3: Patients and doctors can view complete consultation history in chronological order, with AI-recorded and doctor-verified notes saved securely.

FE-2.4: Patient profile shows scan history, past reports, and AI-generated doctor notes.

1.7.3 Module 3: Appointment & Schedule Management

FE-3.1: Patients book physical or virtual appointments with selected specialists

FE-3.2: Admin can view, edit, or delete any appointment

FE-3.3: Doctors view upcoming appointments and patient profiles

FE-3.4: Appointment rescheduling, cancellation, and confirmation notifications.

FE-3.5: View appointment history and real-time status.

1.7.4 Module 4: Diagnostic & Scan Management

- FE-4.1:** Doctors recommends scans/tests during consultations
- FE-4.2:** Patients upload external scans via portal or app
- FE-4.3:** Admin uploads scans for in-hospital tests
- FE-4.4:** All scans linked to patient ID and accessible by assigned doctors
- FE-4.5:** Secure upload with format/size validation and preview

1.7.5 Module 5: AI Diagnostic Engine

- FE-5.1:** Doctor presses a “Start Diagnosis” button after uploading a scan or report.
- FE-5.2:** System auto-selects the relevant model (MRI, Retinal or Textual)
- FE-5.3:** Brain MRI model analyzes tumor, Alzheimer’s
- FE-5.4:** Retinal model analyzes diabetic retinopathy, glaucoma, AMD, hypertensive retinopathy
- FE-5.5: Doctors view AI-generated insights based on the input**
- FE-5.6:** Doctors can edit, verify, or reject the model output
- FE-5.7:** Diagnostic output linked to patient report for further review

1.7.6 Module 6: Explainable AI & Transparency

- FE-6.1:** Visual maps (GradCAM, SHAP) highlight important image regions
- FE-6.2:** Short text explanations of why the AI focused on specific areas
- FE-6.3:** Helps doctors approve or modify results confidently
- FE-6.4:** All explainability data saved with report history

1.7.7 Module 7: Final Report & Summary Generation

- FE-7.1:** Doctor confirms or modifies AI-generated diagnosis
- FE-7.2:** Final report includes both AI and doctor inputs
- FE-7.3:** System auto-generates a simplified summary for patients
- FE-7.4:** Report and summary uploaded to patient portal

1.7.8 Module 8: Virtual Consultation Management

- FE-8.1:** Secure video link/chat room generated for online appointments
- FE-8.2:** Logs doctor-patient discussion notes during virtual consultation
- FE-8.3:** Real-time note exchange in a shared comment section
- FE-8.4:** Comments saved per appointment and linked to report

1.7.9 Module 9: Payment Gateway Integration

- FE-9.1:** Show consultation fee before booking
- FE-9.2:** Support multiple payment modes (cards, wallets, transfers)
- FE-9.3:** Secure transaction and confirmation invoice

FE-9.4: Patient can view payment history in dashboard

1.7.10 Module 10: Multi-language Support

FE-10.1: Interface available in English + Urdu

FE-10.2: Patient reports and summaries translated for easy understanding

FE-10.3: Language toggle in settings

FE-10.4: Doctor notes stay in English for consistency

1.7.11 Module 11: Feedback & Continuous Learning Module

FE-11.1: Doctors can flag incorrect or low-confidence AI predictions (e.g., false positives/negatives)

FE-11.2: Flagged cases (including scans/reports, AI output, and doctor comments) are stored in a structured feedback database

FE-11.3: Admins can review flagged cases and validate, correct, or annotate them with ground truth

FE-11.4: The system groups flagged entries and visualizes recurring error patterns by disease or modality

FE-11.5: Analytics dashboard highlights common misclassifications and feedback frequency for quality monitoring

FE-11.6: Verified feedback cases can be exported as a curated dataset for future retraining of AI models

1.7.12 Module 12: Accessibility & Assistive Tools

FE-12.1: Enable screen reader support across all patient interfaces

FE-12.2: Add text-to-speech for diagnostic summaries and doctor messages

FE-12.3: Provide high-contrast and large-text display options in user profile settings

FE-12.4: Voice command support for key actions (e.g., “Book Appointment”, “Read Report”)

1.8 Related System Analysis/Literature Review

1.7.1 Literature Review

Previous research in AI-assisted medical diagnostics and healthcare automation has focused primarily on developing disease-specific detection systems. Studies in medical imaging (e.g., MRI, CT, and retinal scan analysis) demonstrate that artificial intelligence can effectively identify early disease indicators with high accuracy (Kermany et al., 2018; Pereira et al., 2016). However, most

of these systems operate within narrow clinical domains and lack integration between multiple diagnostic sources, such as brain imaging, and retinal data.

Existing AI-based diagnostic tools such as the RAIDS (Retinal AI Diagnosis System), BTDS (Brain Tumor Detection System), and MediScanAI illustrate these limitations.

- RAIDS focuses exclusively on retinal disease detection only.
- BTDS specializes in MRI-based tumor analysis but does not extend to other neurological or retinal conditions.
- MediScanAI, while capable of textual report interpretation, lacks image-processing capabilities and cross-modal analysis.

Most available systems also fail to provide a centralized medical workflow, where doctors and patients can interact seamlessly within the same platform. These tools often require manual data transfer between diagnostic models, medical professionals, and patients, reducing efficiency and increasing the risk of human error.

Recent advancements in Explainable Artificial Intelligence (XAI) and deep learning interpretability tools (e.g., GradCAM and SHAP) have emphasized the need for transparent AI decision-making in medical contexts. Research indicates that interpretability is crucial for clinical adoption, as it allows healthcare professionals to validate AI outputs and build trust in automated systems (Tjoa & Guan, 2020). Furthermore, integration of speech recognition and automated transcription technologies in medical consultations has shown promise in improving documentation accuracy and reducing physician workload.

The proposed **MediFusion Vision** system directly addresses these gaps by introducing a multi-modal AI diagnostic framework that unifies medical imaging and consultation management in a single platform. It combines AI-powered disease detection, explainable visualization tools, and an AI note-taking assistant to create a transparent, collaborative, and efficient diagnostic ecosystem. By merging deep learning, explainable AI, and real-time consultation tools into one solution, MediFusion Vision contributes a comprehensive and scalable model for next-generation intelligent healthcare systems.

1.7.2 Related System Analysis

Table 1.1 Related System Analysis with proposed project solution

Application Name	Weakness	Proposed Project Solution
RAIDS	Only handles eye diseases via retinal images; no support for other inputs	Supports retina-based diagnosis alongside brain, chest, and lab report analysis
BTDS	Only detects brain tumors; lacks support for other brain diseases or modalities	Covers multiple brain-related diseases and integrates with chest, eye, and text data

1.9 Tools and Technologies

Table 1.2 Tools and Technologies for Proposed Project

Tools And Technologies	Tools	Version	Rationale
	Visual Studio Code	2024	IDE
	Android Studio	2024	For Android app testing/emulation
	Figma	CSC 6	Design Work
	Docker	24.x	Deployment & Containerization
	MS Word	2021	Documentation and report writing
	Python	3.123	AI Model Development
	Pytorch	2.15	Analyze MRIs, retinal images
	Jitsi Meet API	Web-based	Virtual Video Consultations
	Git	2.43	Version Control
	Open AI	Chatgpt-4	AI-powered conversational assistant for parents and basic sign communication
	SHAP/GradCAM	Latest	Explainable AI Visuals
	React JS	17	Web Development
	PostgreSQL	Latest	DBMS

	Flutter	0.74+	Cross-platform mobile development framework
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1.10 Project Contribution

The **MediFusion Vision** system introduces several technical and conceptual contributions that distinguish it from existing AI-based healthcare and diagnostic platforms.

It integrates artificial intelligence, multimodal medical data processing, and real-time consultation management into a unified web and mobile solution designed for patients, doctors, and administrators within a secure and explainable diagnostic environment.

- **AI-Powered Multi-Disease Detection:**

Utilizes advanced deep learning models to analyze brain MRI scans and retinal images. The system can detect multiple conditions such as Brain Tumor, Stroke, Alzheimer's, Diabetic Retinopathy, Glaucoma, and AMD with improved accuracy and efficiency.

- **Explainable AI Visualization:**

Integrates interpretability tools such as GradCAM and SHAP to visually explain AI predictions. This transparency helps doctors validate diagnostic outcomes and enhances clinical trust in AI-assisted decisions.

- **AI Consultation Note-Taker:**

Employs speech recognition to automatically capture consultation details, summarizing key medical observations. This reduces manual documentation effort for doctors and ensures comprehensive, accurate consultation records.

- **Unified Diagnostic Platform:**

Combines multimodal diagnostic capabilities (image, text, and report analysis) into one integrated system accessible through both web and mobile applications, reducing dependency on multiple separate tools.

- **Role-Based Access Management:**

Implements secure access control with dedicated dashboards for patients, doctors, and administrators, ensuring data confidentiality and regulatory compliance within healthcare standards.

- **Comprehensive Diagnostic Reporting:**

Generates AI-assisted diagnostic summaries verified by medical professionals, producing reports that combine machine intelligence and human expertise for higher reliability.

- **Continuous Learning and Feedback Loop:**

Supports an automated system feedback mechanism that collects post-diagnosis validation data from doctors to progressively enhance AI performance through self-learning and retraining.

Contribution Impact:

By combining artificial intelligence, explainable visualizations, multimodal diagnostics, and

automation, MediFusion Vision transforms traditional healthcare workflows into an intelligent, transparent, and patient-centered diagnostic experience.

These contributions improve diagnostic speed, accuracy, and collaboration between patients and medical professionals, setting a strong foundation for the future of AI-assisted healthcare systems.

1.11 Relevance to Course Modules

- **Software Engineering:** Applied all major SDLC phases including requirement analysis, system design, implementation, and testing to ensure structured and maintainable system development.
- **Artificial Intelligence & Machine Learning:** Implemented deep learning models for disease detection and classification using MRI and retinal medical data. Integrated Explainable AI (GradCAM and SHAP) for visual interpretation and validation of diagnostic results.
- **Web & Mobile Application Development:** Developed web application using React.js and Flutter for the mobile application, following responsive design and cross-platform UI principles for seamless user experience across devices.
- **Database Systems:** Utilized PostgreSQL for secure storage and retrieval of patient records, doctor data, appointment details, and diagnostic reports, ensuring relational integrity and optimized data management.
- **Human-Computer Interaction (HCI):** Designed patient-friendly and accessible user interfaces with features such as Urdu translation, voice assistance, and simplified navigation to accommodate users with varying technical proficiency.
- **Computer Networks & Cloud Computing:** Implemented secure, real-time data communication for virtual consultations and AI-based report synchronization through cloud deployment and encrypted network connections.
- **Data Science:** Applied preprocessing, normalization, and data augmentation techniques on multimodal medical datasets to improve model performance and ensure diagnostic reliability.

2 Chapter 2: Requirement Analysis

This chapter presents the requirement analysis for the MediFusion Vision system. It defines the system's primary user classes, their interactions with the web and mobile applications, and the core functional requirements derived from AI-assisted diagnostic modeling and prototype-based design. The analysis focuses on how patients, doctors, and administrators utilize key modules such as AI diagnosis, consultation management, medical report generation, and system monitoring to ensure accurate, transparent, and efficient healthcare delivery.

2.1 User Classes and Characteristics

The MediFusion Vision system serves multiple user classes, each with specific goals, roles, and interaction patterns. These include Patients, Doctors, and an Administrator. Their characteristics, technical proficiency, frequency of system use, and privileges are described below.

Table 2-1: User Classes of the Proposed System

User Class	Description
Patient	Patients are the primary end-users who use MediFusion Vision to access healthcare services through AI-assisted diagnostics. They can register, log in, search for doctors by specialization, and schedule either virtual or in-person consultations. Patients can upload external medical scans, view diagnostic reports, download or share results, and make secure online payments. The interface is designed for simplicity and accessibility, supporting Urdu translation and voice-based navigation to assist users with limited technical experience. • Technical Proficiency: Basic to moderate (general smartphone or web users). • Interaction Frequency: Occasional to frequent, depending on medical needs. • Access Mode: Web and mobile application. • Privileges: Can register, log in, update profile, search doctors, book/reschedule/cancel appointments, upload scans, make payments, view diagnostic reports, and submit feedback or complaints.
Doctor	Doctors are verified medical professionals who utilize MediFusion Vision to review AI-generated diagnostic outcomes, conduct consultations, and finalize patient reports. They access detailed patient histories, view consultation schedules, and interact

	with the AI diagnostic system to verify findings. The AI Note-Taker automatically records consultation details, reducing manual entry. Doctors can view Explainable AI visuals (GradCAM and SHAP) to ensure diagnostic transparency and reliability before confirming reports. • Technical Proficiency: High (trained professionals with experience in digital tools). • Interaction Frequency: Daily, during work shifts or active consultation hours. • Access Mode: Web and mobile dashboards. • Privileges: Can log in, manage appointments, view patient data, conduct consultations, use AI diagnostic tools, review and verify reports, communicate with patients, and provide medical feedback.
Administrator	The Administrator manages and monitors the overall MediFusion Vision platform. Admins approve or reject doctor registrations, oversee user management, monitor appointments, and ensure proper system operation. They are responsible for uploading hospital-generated scans, maintaining audit logs, handling user feedback, and generating performance and compliance reports. The Admin ensures that all AI, data, and consultation modules function securely and smoothly across the system. • Technical Proficiency: Experienced system operator or healthcare administrator. • Interaction Frequency: Daily, for system monitoring and administrative management. • Access Mode: Web-based administrative dashboard (desktop preferred). • Privileges: Can approve/reject doctor accounts, manage users, monitor appointments, upload hospital scans, maintain audit logs, handle feedback and complaints, and generate system performance insights.

2.1.1 Use case Diagram

The following Use Case Diagrams illustrate the interactions between the main users of the MediFusion Vision system and its core functionalities. Each diagram represents how patients, doctors, and the administrator interact with different modules to perform diagnostic, consultation,

and administrative tasks, ensuring accurate medical evaluation and efficient healthcare management.

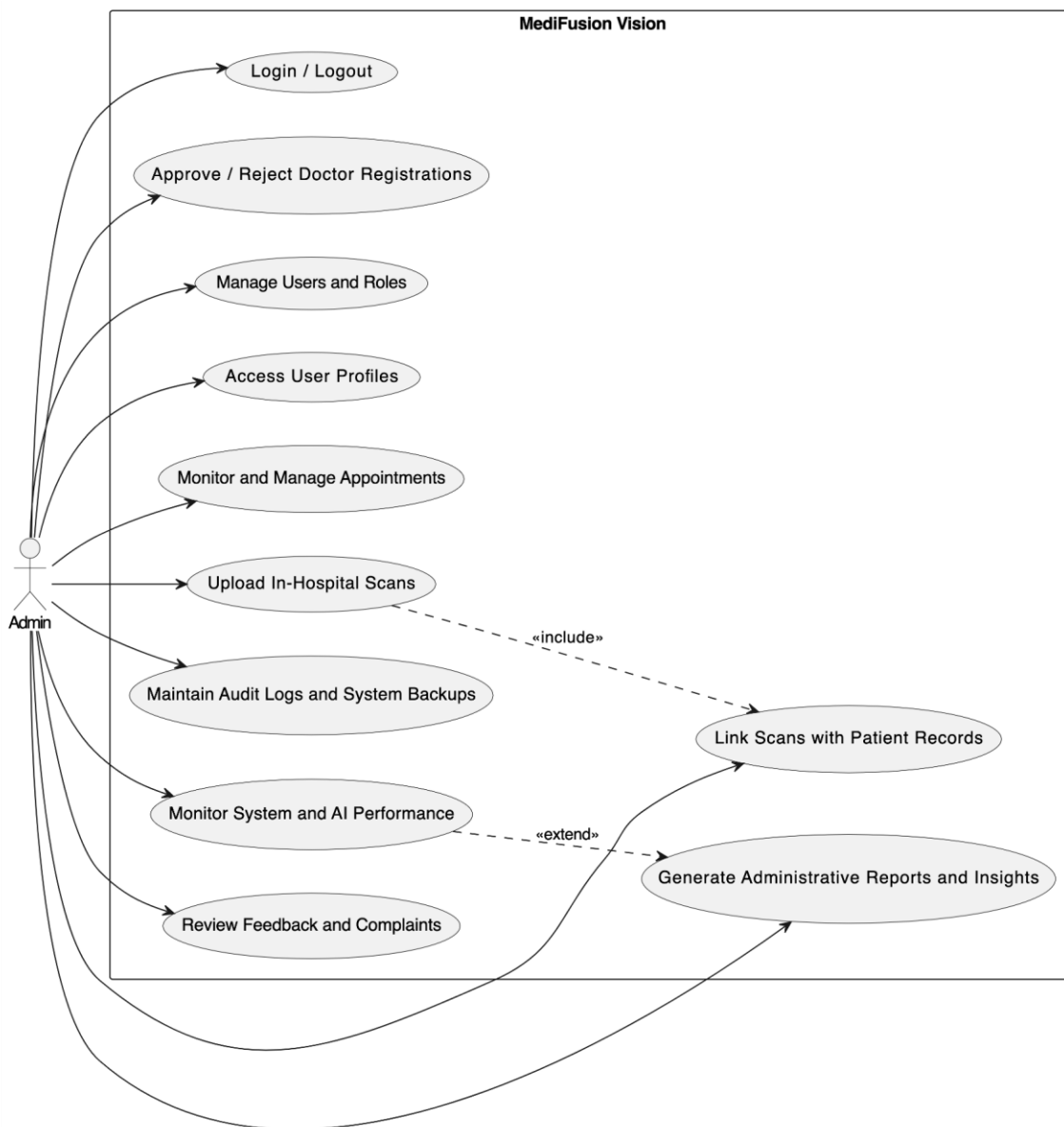


Figure 2-1: Use Case Diagram for Admin

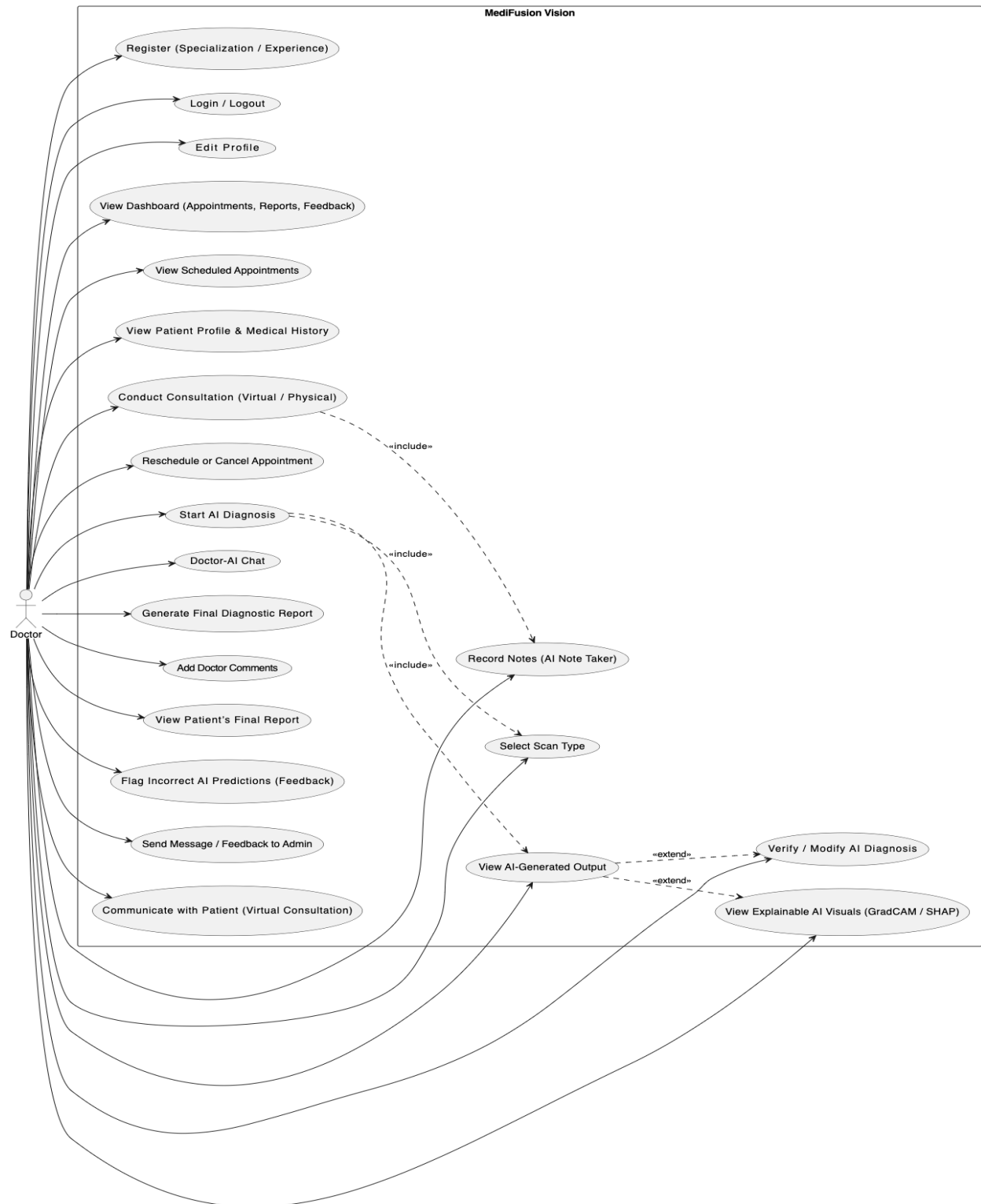


Figure 2-2: Use Case Diagram for Doctor

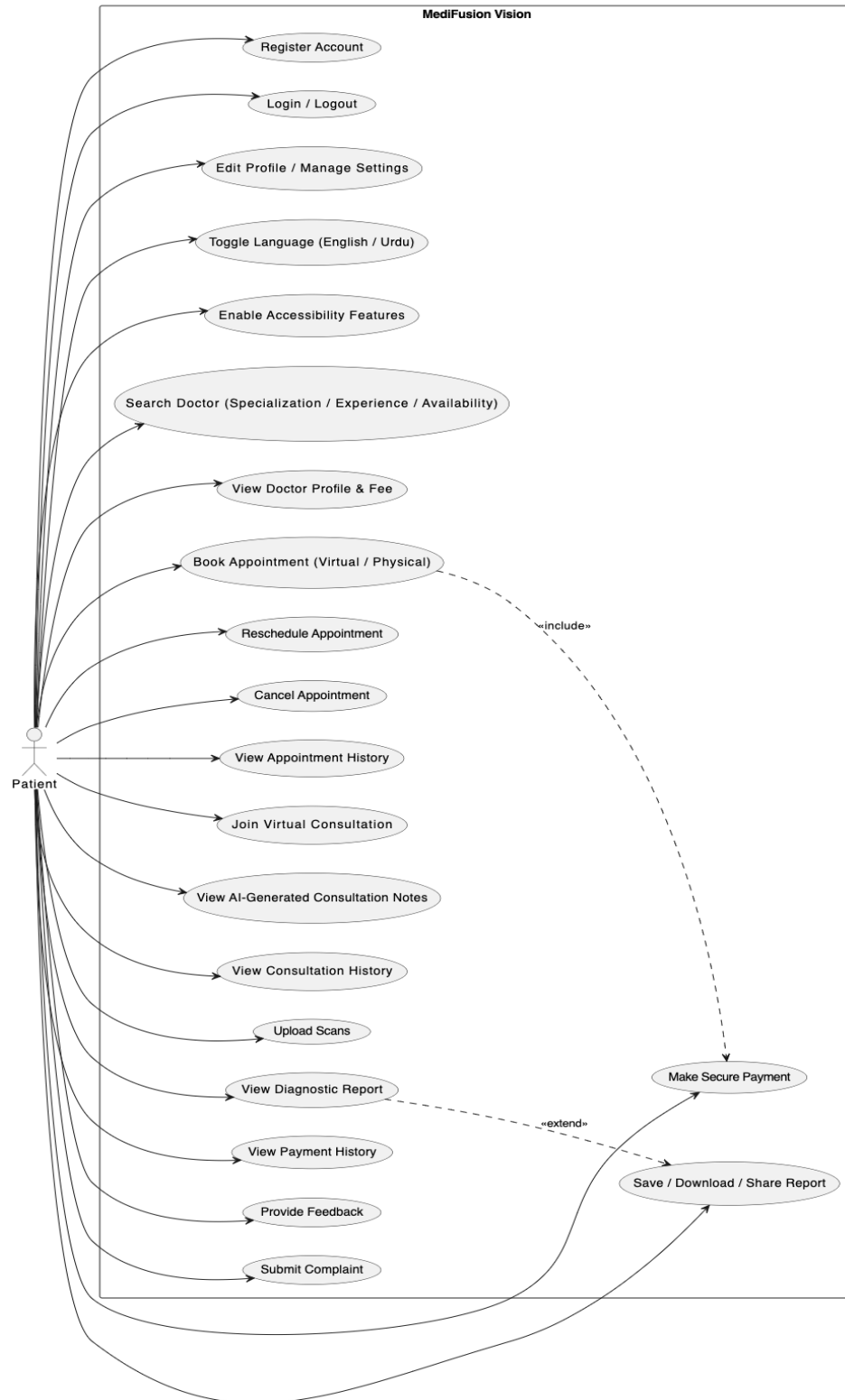


Figure 2-3: Use Case Diagram for Patient

2.1.2 Requirement Identifying Technique

This section describes the techniques used to identify and document the functional requirements of the *MediFusion Vision* system.

For this project, a **Mockup-Based Requirement Analysis** approach is used. This method was selected because the system involves multiple user roles that are **Admin, Doctor, and Patient** interacting through distinct dashboards and medical diagnostic interfaces. Using mockups enables a clear understanding of how each user will interact with the system and allows functional requirements to be derived directly from visible interface behaviors.

1. Mockup-Based Requirement Analysis

This technique captures all functional requirements triggered by user interactions.

User interface mockups were created to visualize how doctors, patients, and admins perform their respective tasks, such as uploading scans, running diagnostic models, booking consultations, and generating reports.

Each screen was assigned a **unique Mockup ID** (e.g., M1 for Login Page, M2 for Doctor Dashboard, M3 for Patient Portal, M4 for AI Diagnosis Screen, M5 for Final Report Page).

Every interactive element—buttons, forms, menus, upload fields, and charts—was examined, and a corresponding **Functional Requirement (FR)** was derived, describing the system’s expected response.

Steps Followed

1. Identify all major system features and group them into separate mockup screens.
2. Assign a unique Mockup ID to each screen.
3. Create interactive mockups using Figma to represent real-world workflows.
4. Identify all actionable UI components on each mockup (e.g., “Upload Scan,” “Run AI Model,” “Book Appointment”).
5. Derive a testable functional requirement for each interaction describing what the system should do.
6. Document all requirements in a structured table (Section 2.2) linking each FR to its mockup ID and element.

This process ensures **complete alignment between user interface design and functional behavior**, maintaining traceability between visual design elements and their underlying system functionality

2.1.3 Backend Process Requirements (Event–Response Table Technique)

While mockups capture the front-end interactions of the MediFusion Vision, many critical operations take place entirely in the backend. These include user authentication, doctor verification, medical scan validation, AI-driven diagnosis, consultation note generation, report creation, and model retraining based on flagged cases. To accurately represent these system-driven workflows, the Event–Response Table Technique was applied.

This technique identifies key backend events, the system state during each event, and the automated responses triggered by the system. It ensures that real-time and background processes such as AI inference, data storage, and automated notifications, are clearly defined during the requirements phase without delving into implementation details.

Steps Followed:

- Identify backend or automated events such as doctor approval, scan uploads, consultation sessions, AI diagnosis execution, and model retraining triggers.
- Define the system state during each event.
- Specify the automated system response, including data validation, AI processing, report generation, and user notifications.

Table 2-2: Event Response Table of the Proposed System

Event	System State	Response / System Behavior
Patient registers	Registration service active	1. Store patient details in database. 2. Send confirmation message/email.
Doctor registers with qualifications	Registration service active	1. Save doctor data and qualifications. 2. Mark status as <i>Pending Approval</i> .
Admin reviews doctor registration.	Doctor marked as <i>Pending Approval</i> .	1. Approve or reject doctor. 2. Update status accordingly and notify doctor
Patient logs in	Valid credentials provided	Authenticate and redirect to doctor dashboard.
Doctor logs in	Valid credentials provided	Authenticate and redirect to doctor dashboard.
Patient books appointment (virtual or physical)	Doctor available and slot open.	1. Confirm appointment. 2. Update both doctor and patient schedules.
Patient or Doctor cancels or	Appointment exists.	1. Update appointment status in database. 2. Notify all concerned users (patient, doctor, and admin).

reschedules appointment		
Patient uploads an image file for diagnosis	Patient authenticated; upload service active.	1. Validate file type and size. 2. If file fails validation → reject upload, log error, and return “Invalid file format or size” to patient. 3. If validation passes → store image securely in database/cloud storage. 4. Update patient record status to “Uploaded (Awaiting Consultation)”
Doctor starts consultation session	Patient scans available; doctor authenticated.	1. Activate AI Note Taker to transcribe consultation in real-time. 2. Load patient’s uploaded scans for review. 3. Begin consultation timer/session tracking.
AI Note Taker detects speech or discussion.	Consultation ongoing; AI transcription service running.	1. Continuously transcribe speech into text. 2. Generate consultation notes draft in background. 3. Save progressive transcript to database under patient’s profile
Doctor triggers AI diagnosis on patient scans	Scans validated; AI model idle; consultation active.	1. Send scan to AI model for inference. 2. Generate diagnostic result + explainable visualization (e.g., Grad-CAM heatmap). 3. Store results and confidence score in DB. 4. Display to doctor within session.
Doctor reviews and verifies the AI diagnosis	Consultation active; AI result displayed.	1. Doctor confirms or modifies AI findings. 2. Final verification status saved (“Verified by Doctor”). 3. AI notes and doctor remarks merged.
Final report generation triggered	Doctor verification complete	1. Combine AI results, doctor’s notes, and explainable visuals into one report. 2. Generate PDF with digital signature and timestamp. 3. Store in secure records. 4. Notify patient that the final report is available for download.
Doctor flags a case as ‘AI Incorrect’ (Flagged case).	Doctor verification step	1. Mark record with FlaggedForRetrain + reason + doctor comments. 2. Collect all relevant artifacts (scan, AI output, doctor-corrected label, notes). 3. Add flagged-cases dataset queue. 4. Notify ML team/admin dashboard.
Patient requests report download.	Report status is “Verified” and available	1. Retrieve report from storage. 2. Generate PDF and audit trail entry. 3. Send secure download link via email.

2.2 Functional Requirements

This section describes the functional requirements of the MediFusion Vision system, derived from user interactions and mockups. To ensure clarity and maintain a stable specification, each requirement is documented at the feature level. Each functional requirement includes:

- A unique identifier (e.g., FR-1.1, FR-3.2).
- A clear and testable requirement statement.
- A reference to the originating mockup screen.
- An optional Business Rule, if the behavior is governed by a domain-specific policy, condition, or constraint.

2.2.1 Recommended Format:

This subsection presents the standard format used to record and organize all functional requirements identified through the Mockup-Based Requirement Analysis. Each requirement is derived from specific UI elements or interactions and is documented in a consistent, tabular form to ensure traceability, clarity, and ease of verification.

M1-Login Screen

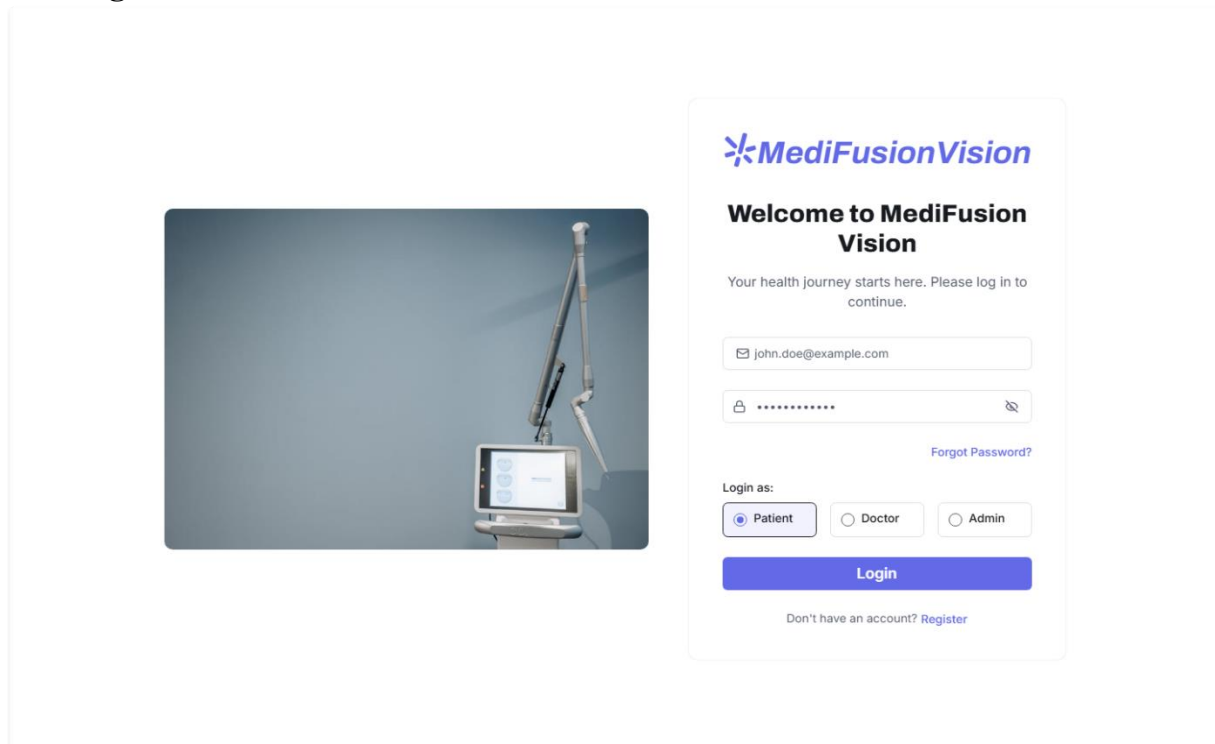


Figure 2-4: Login Page

Table 2-3: Functional Requirements derived from Mockup M1

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Email input field	FR1.1 The system shall allow users to enter their registered email address to log in.	Email must be in a valid format and already registered.
Password input field	FR1.2 The system shall allow users to enter their password to log in.	Password must match the email address provided.
Password visibility toggle	FR1.3 The system shall allow users to toggle the visibility of the entered password.	For user convenience only; visibility does not affect authentication.
Login as radio buttons	FR1.4 The system shall allow users to select their login type: Patient, Doctor, or Admin.	User role determines dashboard and access rights.
Login button	FR1.5 The system shall authenticate users and redirect them to their respective dashboards upon successful login.	Only active, verified accounts can log in.
Register link	FR1.6 The system shall provide a link to the registration page for new account creation.	Only for users who do not have an account yet.
Forgot password link	FR1.7 The system shall provide a link for users to initiate the password recovery process.	Password reset process must verify user identity.

M2-Dashboard Screen (Doctor Web)

The Dashboard screen provides doctors an at-a-glance overview of their upcoming appointments, recent AI diagnostic results, and pending medical reports for review. It allows navigation to detailed views and one-click actions for core clinical tasks.

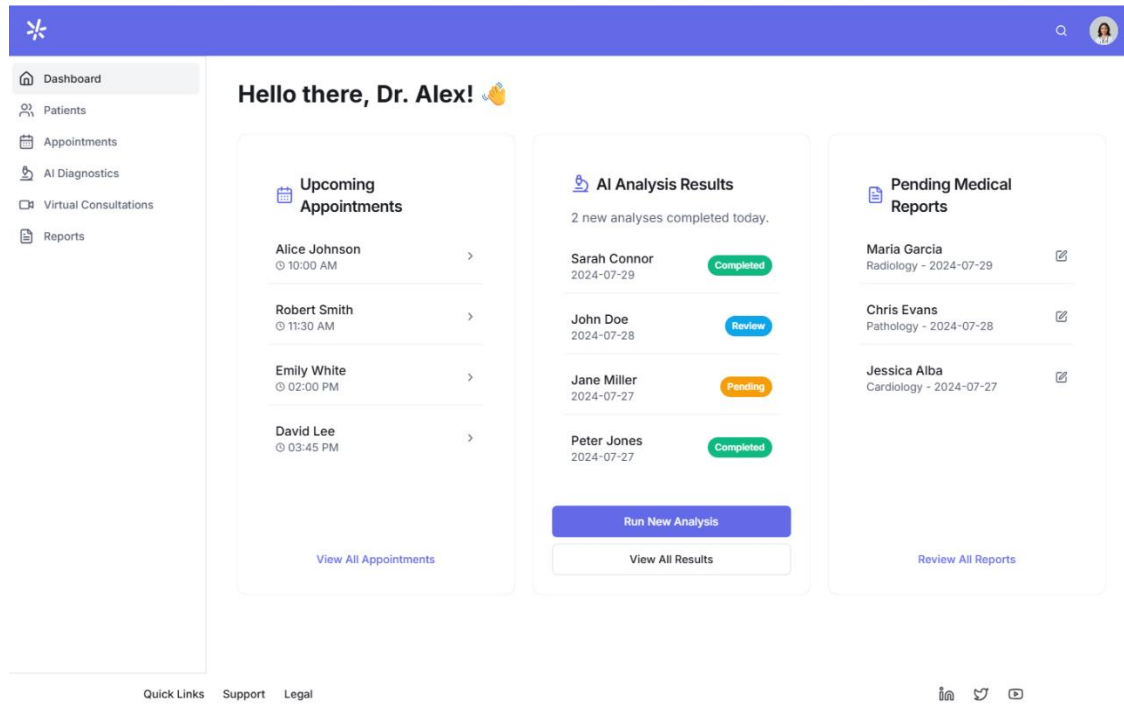


Figure 2-6: Dashboard Screen – Doctor Web

Table 2-4: Functional Requirements derived from Mockup M2

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Upcoming Appointments list	FR2.1 The system shall display the doctor's scheduled upcoming appointments with basic patient details.	Only appointments assigned to the logged-in doctor are shown.
View All Appointments link	FR2.2 The system shall provide a link to the full appointments calendar for the doctor.	Link accessible only to authorized users.
AI Analysis Results panel	FR2.3 The system shall summarize recent AI analyses performed for the doctor's patients.	Only results for patients under this doctor's care are shown.
Run New Analysis button	FR2.4 The system shall allow the doctor to initiate a new AI diagnostic workflow.	Only accessible to doctors with required permissions.
View All Results button	FR2.5 The system shall provide access to a complete list of AI analysis results.	Accessible to doctors for their own patients only.
Pending Medical Reports list	FR2.6 The system shall display a list of medical reports needing the doctor's review.	Only reports assigned for the logged-in doctor's review.

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Review All Reports link	FR2.7 The system shall provide navigation to a page for reviewing all pending reports.	Only accessible to authorized users.

M3-AI Diagnostic Workflow Screen

This screen allows doctors to upload patient medical scans, run AI-powered diagnostic analysis, view results and insights, and review patient details. It streamlines the AI diagnostics process for clinical decision support.

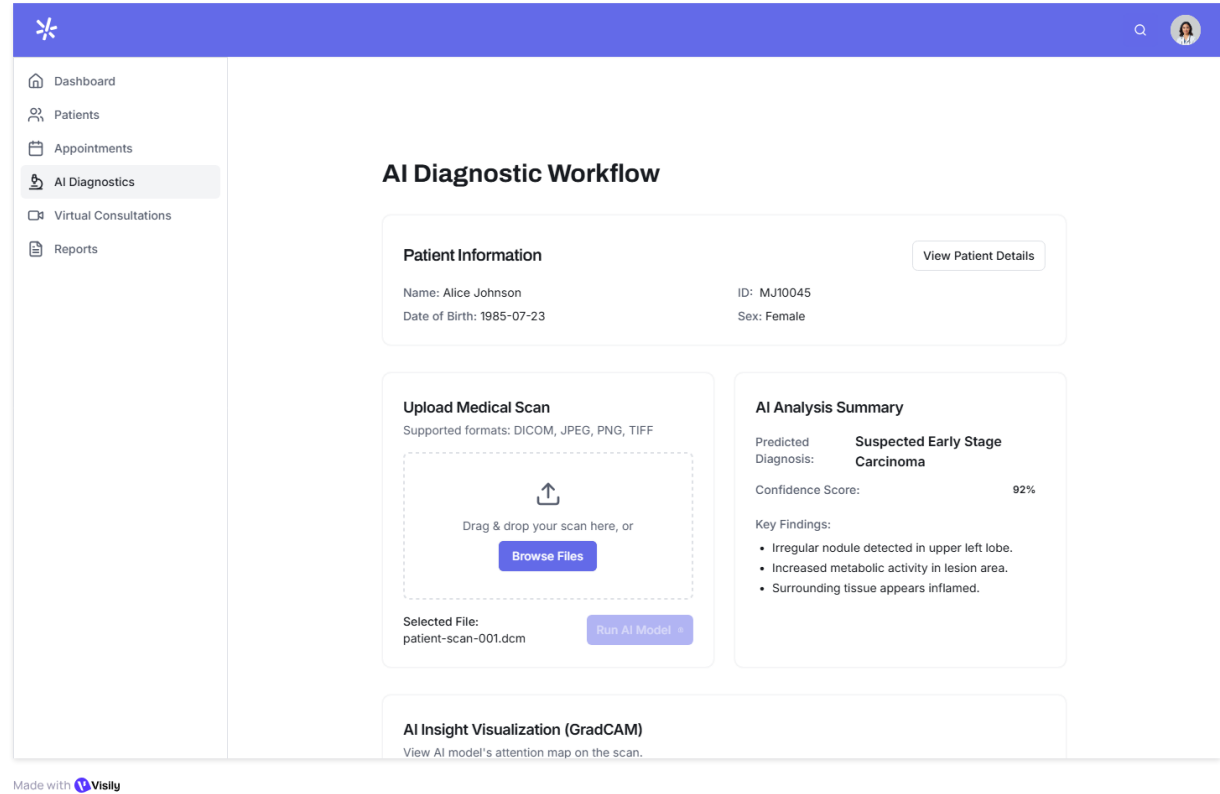


Figure 2-7: AI Diagnostic Workflow Screen

Table 2-5: Functional Requirements derived from Mockup M3

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Patient Information panel	FR3.1 The system shall display patient demographic and identifier details for reference.	Only accessible for authorized patients.
View Patient Details button	FR3.2 The system shall allow navigation to a detailed patient profile view.	Profile access follows hospital data privacy policies.

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Upload Medical Scan (drag/drop & browse)	FR3.3 The system shall allow doctors to upload medical scans in supported formats.	Only DICOM, JPEG, PNG, or TIFF formats allowed.
Run AI Model button	FR3.4 The system shall start the AI diagnostic analysis on the uploaded scan.	Button enabled only when a valid scan is selected.
AI Analysis Summary panel	FR3.5 The system shall display the predicted diagnosis, confidence score, and key findings from AI.	Results shown only after completion of analysis.
AI Insight Visualization section	FR3.6 The system shall show AI visualizations (e.g., GradCAM heatmaps) to explain model attention.	Available only for compatible scan types.

M4-Appointment Management Screen

Doctors manage appointments using a calendar interface, which shows all scheduled, upcoming, and past appointments. The screen allows adding new appointments, switching calendar views, and viewing appointment details.

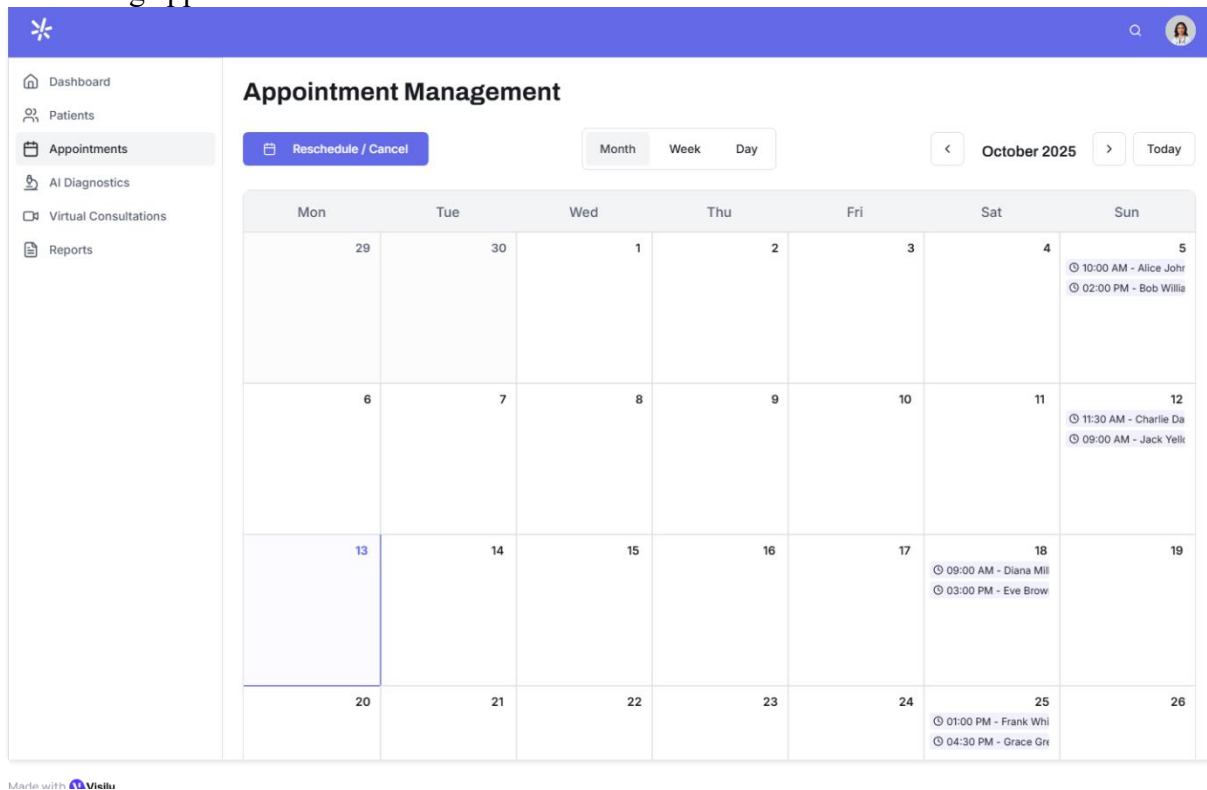


Figure 2-8: Appointment Management Screen

Table 2-6: Functional Requirements derived from Mockup M4

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Add New Appointment button	FR4.1 The system shall allow doctors to schedule new patient appointments via a form.	Only doctors with scheduling privileges may use.
Calendar view (Month/Week/Day toggle)	FR4.2 The system shall let users switch between monthly, weekly, and daily calendar views.	Default view is Month on page load.
Appointment entries on calendar	FR4.3 The system shall display individual appointment details as entries on the calendar.	Only appointments for the logged-in doctor appear.
Navigate month/week/day and 'Today'	FR4.4 The system shall allow users to navigate to different dates using arrows and the Today button.	Cannot view appointments beyond their assigned role/date scope.

M5-AI Note Taker Screen

On this screen, the system auto-generates consultation notes from patient-doctor dialogue, and allows doctors to add edits or corrections. This helps in maintaining accurate, comprehensive visit records efficiently.

MediAI

Dashboard Patient Records **Consultations** Analytics Settings Support

Search patient records or medi

AI Note Taker

AI Status: Live Manual Entry ☒ Auto Summary

AI-Generated Consultation Notes

[00:00] Dr. Reed: Good morning, Maria. Thanks for coming in. What brings you here today? [00:05] Patient: Good morning, Doctor. I've been having this persistent dull ache in my lower back for about three weeks now. It's worse when I stand for long periods or lift things. [00:15] Dr. Reed: I see. And on a scale of one to ten, with ten being the worst, how would you rate the pain? [00:19] Patient: It's usually around a four, but sometimes it flares up to a six or seven, especially by the end of the day. [00:25] Dr. Reed: Has it radiated down your legs at all, or do you have any numbness or tingling? [00:29] Patient: No, it stays localized in my lower back. No numbness or tingling, thankfully. [00:35] Dr. Reed: Any recent injuries, falls, or changes in your activity level that might have triggered this? [00:40] Patient: Not really. I work at a desk most days, but I try to get some walks in. I haven't done any heavy lifting lately. [00:48] Dr. Reed: Are you experiencing any other symptoms, like fever, chills, or changes in bowel or bladder habits? [00:54] Patient: No, none of those. I feel generally well, just this nagging back

Doctor's Corrections / Additions

Initial assessment complete. Patient presents with mechanical lower back pain. Advised conservative management with heat/ice, OTC NSAIDs, gentle stretching, and activity modification. Will provide exercise handout. Follow-up in 2 weeks if no significant improvement or worsening of symptoms. No red flags identified.

Save Notes Attach to Report Discard

Patient Details
Name: Maria Rodriguez
ID: P-456789
DOB:

Consultation Details
Session ID: CS-20240726-001
Timestamp: July 26, 2024, 10:30 AM

Figure 2-9: AI Note Taker Screen

Table 2-7: Functional Requirements derived from Mockup M5

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
AI-Generated Consultation Notes	FR5.1 The system shall transcribe and display consultation dialogue as structured notes using AI.	Only available for active, ongoing consultations.
Doctor's Corrections/Additions editor	FR5.2 The system shall allow the doctor to edit or append to the AI-generated notes.	Changes must be saved for inclusion in final report.
AI Status toggle (Manual/Auto Summary)	FR5.3 The system shall let the doctor switch between manual note entry and AI-powered auto-summary.	Default is AI Summary but can be changed at any time.
Save Notes, Attach to Report, Discard	FR5.4 The system shall allow doctors to save notes, attach them to reports, or discard unsaved changes.	Attaching is only possible after notes are saved.

M6-Virtual Consultation Screen

This screen supports secure video consultations between doctors and patients. Features include live video, chat, call status, and controls for muting, ending the call, or adjusting the video view.

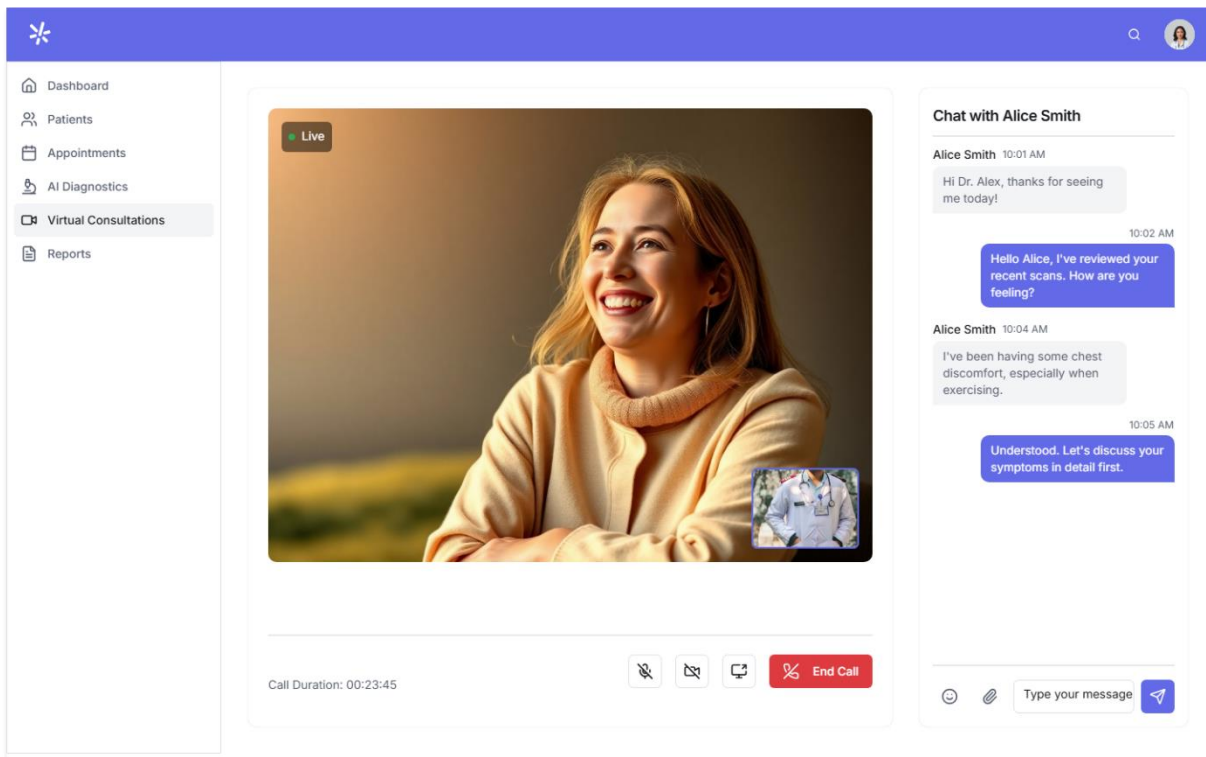


Figure 2-10: Virtual Consultation Screen

Table 2-8: Functional Requirements derived from Mockup M6

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Live video call interface	FR6.1 The system shall provide real-time video and audio communication between doctor and patient.	Only scheduled/authorized consultations.
Call duration display	FR6.2 The system shall track and display current call duration.	Shows duration in hh:mm:ss format.
Mute, Video On/Off, End Call buttons	FR6.3 The system shall allow real-time muting/unmuting, camera toggle, and ending of calls.	Only participants may control their own stream.
Text chat panel	FR6.4 The system shall enable secure text-based messaging during the call.	Messages are accessible only during the call.
Message input field	FR6.5 The system shall provide a field to send messages during the consultation.	Disabled if call is ended/disconnected.

M7-Patient Details Screen

Displays detailed demographic, contact, and medical history information for an individual patient, with quick actions to view reports or start AI diagnostics. For use by doctors and clinical staff.

The screenshot shows a web application interface for patient details. On the left is a sidebar with navigation links: Dashboard, Patients (selected), Appointments, AI Diagnostics, Virtual Consultations, and Reports. The main content area is titled 'Patient Details' and features a card for 'Eleanor Vance' (Patient ID: PV00789). The card displays demographic information: Age (45 years), Gender (Female), and Contact (eleanor.vance@example.com). Below this, there are two buttons: 'View Reports' and 'Initiate AI Diagnostic'. Underneath the card is a 'Medical History' section with two expandable entries. The first entry is 'Diagnosed with Type 2 Diabetes (2020-03-15)', which includes a summary of the diagnosis and treatment. The second entry is 'Hypertension Management (2021-07-22)', which includes a summary of blood pressure management and treatment.

Figure 2-11: Patient Details Screen

Table 2-9: Functional Requirements derived from Mockup M7

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Patient Demographics panel	FR7.1 The system shall display age, gender, contact details, and patient ID.	Data only for the selected/authorized patient.
View Reports button	FR7.2 The system shall provide quick access to all of the patient's diagnostic reports.	Only doctors/authorized staff can access reports.
Initiate AI Diagnostic button	FR7.3 The system shall allow the doctor to start a new AI diagnostic workflow for the patient.	Only accessible if sufficient patient data exists.
Medical History (expandable)	FR7.4 The system shall show the patient's detailed medical history in expandable sections.	Entries updated by authorized medical staff only.

M8-Patient List Screen

Allows clinicians to view, filter, and search all patients assigned to their care, with status indicators, action buttons, and patient summary info in a sortable table.

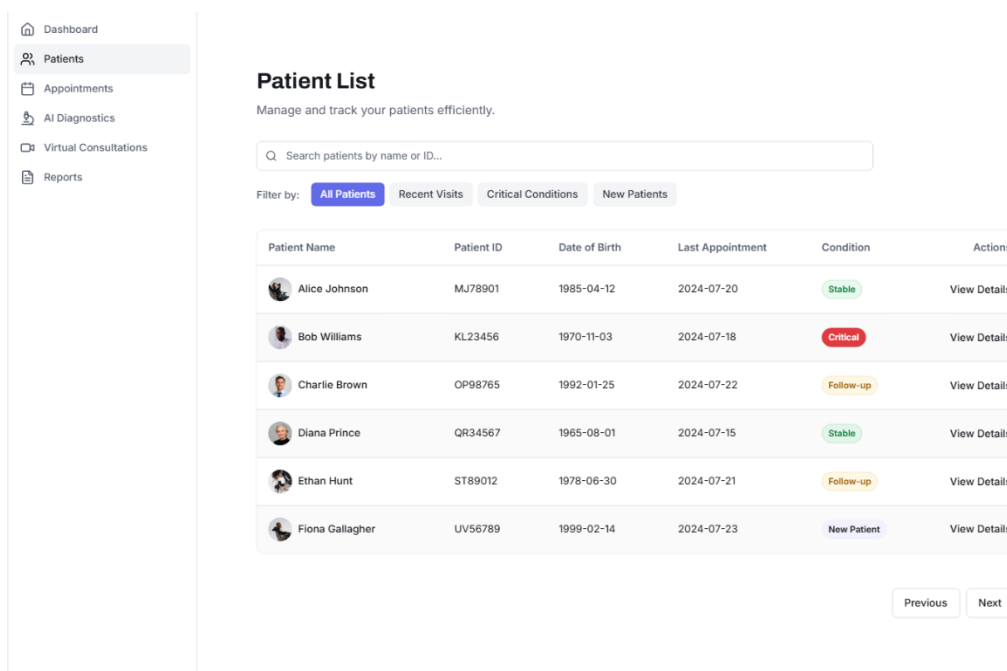


Figure 2-12: Patient List Screen

Table 2-10: Functional Requirements derived from Mockup M8

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Search bar	FR8.1 The system shall allow users to search patients by name or ID.	Search applies only to patients under clinical care.
Filter buttons	FR8.2 The system shall allow filtering by recent visits, critical conditions, and new patients.	Only one filter active at a time.
Patient List table	FR8.3 The system shall show a table with name, ID, date of birth, appointment date, and condition.	Only authorized patients shown.
Add New Patient button	FR8.4 The system shall allow clinicians to register new patients into the system.	Registration requires unique email or ID.
Condition indicator badges	FR8.5 The system shall visually display patient health status (e.g., Stable, Critical).	Condition is updated based on clinical records.
View Details button	FR8.6 The system shall provide navigation to the full details screen for the selected patient.	Only clinicians with clearance can access details.
Pagination controls	FR8.7 The system shall paginate the list, showing navigation for previous and next pages.	Default page size set by system configuration.

M9-Final Diagnostic Report Screen

Presents the finalized, detailed report of a patient's diagnosis, findings, and AI-generated analysis to the doctor for review or sharing. Ensures delivery of accurate results, including radiological and AI-authored sections.

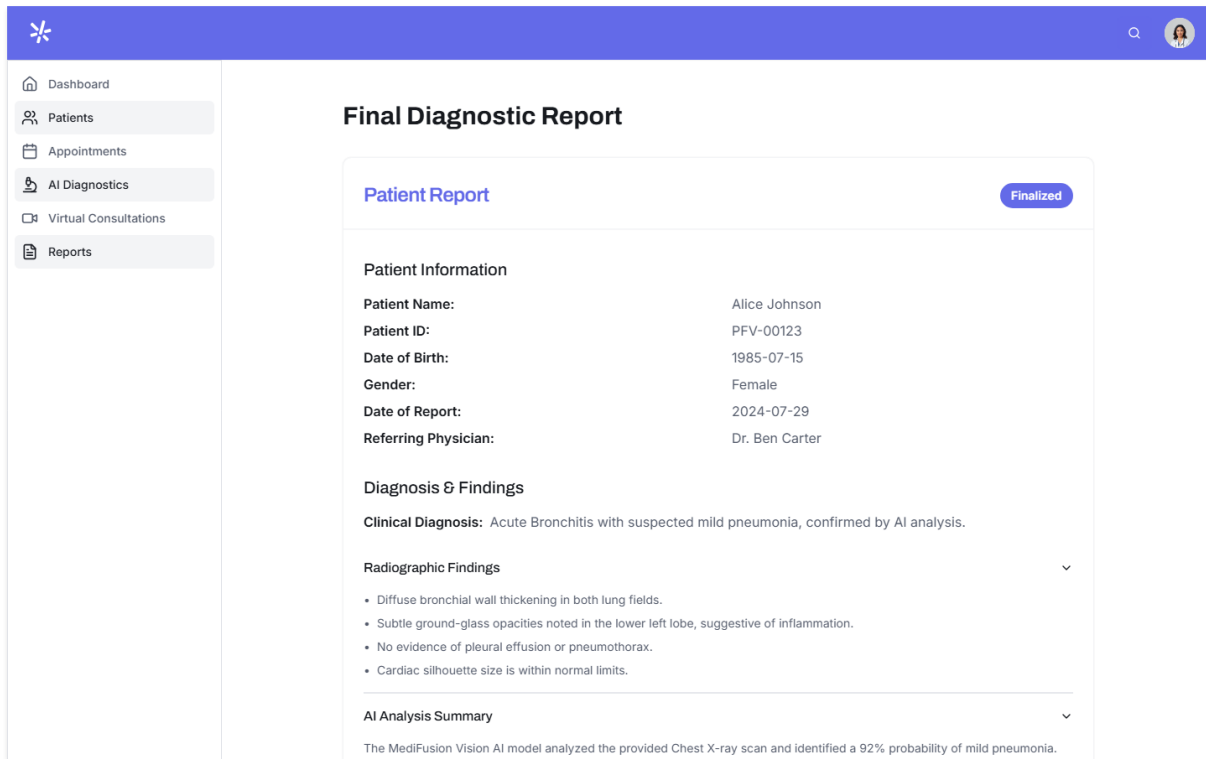


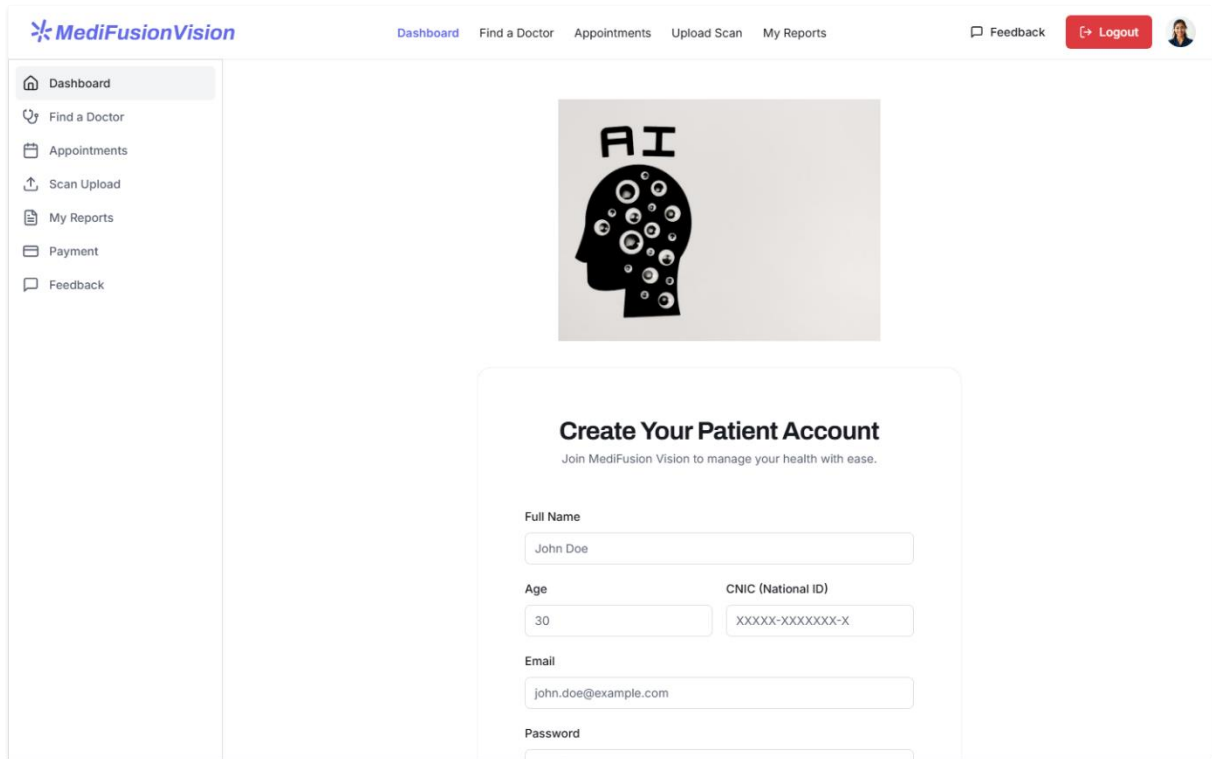
Figure 2-13: Final Diagnostic Report Screen

Table 2-11: Functional Requirements derived from Mockup M9

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Patient Report panel	FR9.1 The system shall display finalized diagnostic reports including patient and physician details.	Only viewable once finalized and by authorized staff.
Diagnosis & Findings section	FR9.2 The system shall display clinical diagnosis and radiographic findings in detail.	Entries are write-protected after finalization.
AI Analysis Summary section	FR9.3 The system shall present AI-generated explanations and probabilities for diagnoses.	AI summary shown only for supported scan/reports.
Finalized status badge	FR9.4 The system shall display a clear indicator when a report is finalized.	Report cannot be edited once finalized.

M10-Patient Registration Screen

This screen enables new patients to create an account by entering personal and contact information (name, age, national ID, email, password).



The image shows a web application interface for MediFusionVision. At the top, there is a navigation bar with the logo on the left and links for Dashboard, Find a Doctor, Appointments, Upload Scan, My Reports, Feedback, and a Logout button. A sidebar on the left contains icons and labels for Dashboard, Find a Doctor, Appointments, Scan Upload, My Reports, Payment, and Feedback. The main content area features a large graphic of a head with 'AI' and a brain-like pattern. Below this is a 'Create Your Patient Account' form with the subtitle 'Join MediFusion Vision to manage your health with ease.' The form includes input fields for Full Name (pre-filled with 'John Doe'), Age (pre-filled with '30'), CNIC (National ID) (pre-filled with 'XXXXX-XXXXXXX-X'), Email (pre-filled with 'john.doe@example.com'), and Password.

Figure 2-14: Patient Registration Screen

Table 2-12: Functional Requirements derived from Mockup M10

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Full Name, Age, CNIC fields	FR10.1 The system shall allow patients to provide their name, age, and national ID during registration.	CNIC must be valid and unique in the system.
Email field	FR10.2 The system shall allow patients to enter an email address for account creation.	Email must be unique and in a valid format.
Password field	FR10.3 The system shall allow patients to set a secure password during registration.	Password must meet system security criteria.
Register button	FR10.4 The system shall create a new patient account upon valid registration submission.	Registration is successful only when all fields are valid.

M11-Patient Dashboard Screen

On login, the patient dashboard displays a welcome message, shortcuts (book appointment, upload scan, view reports), upcoming appointments, and yearly report count.

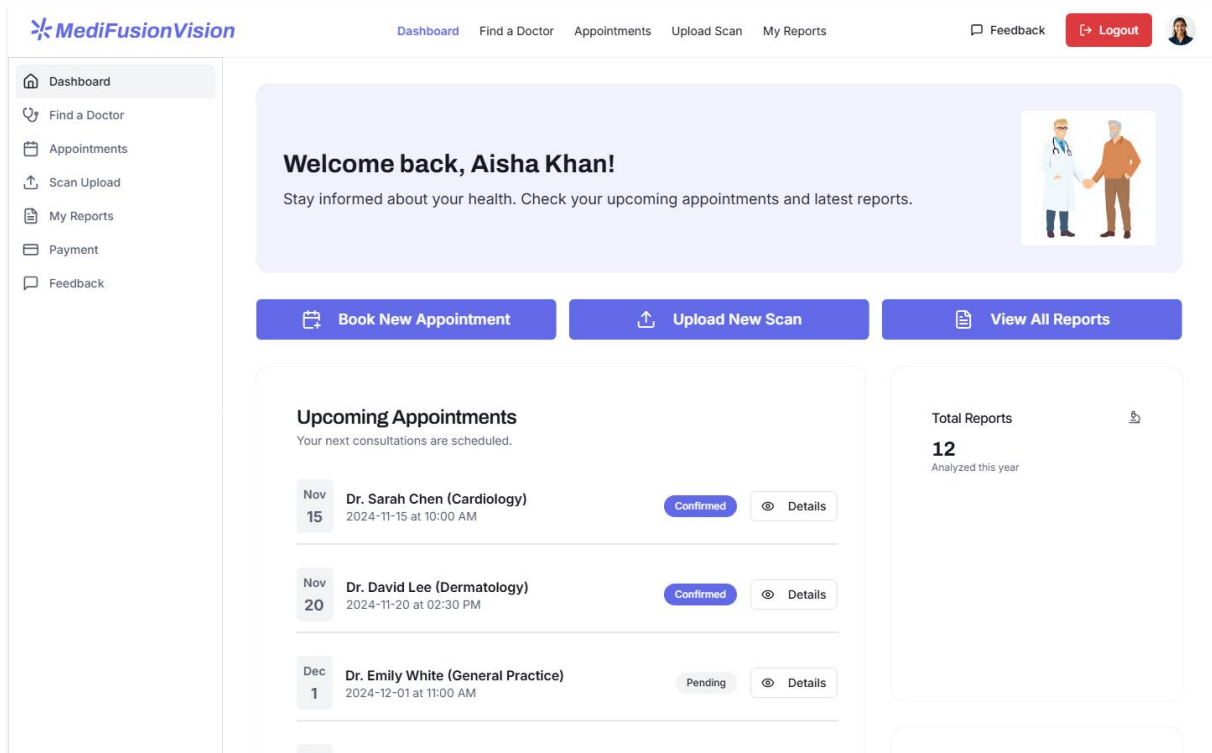


Figure 2-15: Patient Dashboard Screen

Table 2-13: Functional Requirements derived from Mockup M11

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Welcome message	FR11.1 The system shall greet the patient and summarize recent activity on dashboard load.	Only authenticated patients view this content.
Book New Appointment button	FR11.2 The system shall provide a shortcut to initiate appointment booking.	Visible only to active patients.
Upload New Scan button	FR11.3 The system shall permit scan upload access for AI diagnostics.	Upload allowed for verified users only.
View All Reports button	FR11.4 The system shall navigate to the patient's full diagnostic report history.	Access limited to patient's own records.
Upcoming Appointments list	FR11.5 The system shall display all scheduled future consultations for the patient.	Only future appointments for logged-in patient shown.

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Appointment status badges	FR11.6 The system shall indicate status of each appointment (Confirmed/Pending).	Status updated dynamically from backend.
Total Reports indicator	FR11.7 The system shall show the total number of reports for the year.	Count includes only finalized, available reports.

M12-Doctor Search Screen

Patients can search, filter, and browse for doctors using search and specialty filters, and view trending searches and doctor cards.

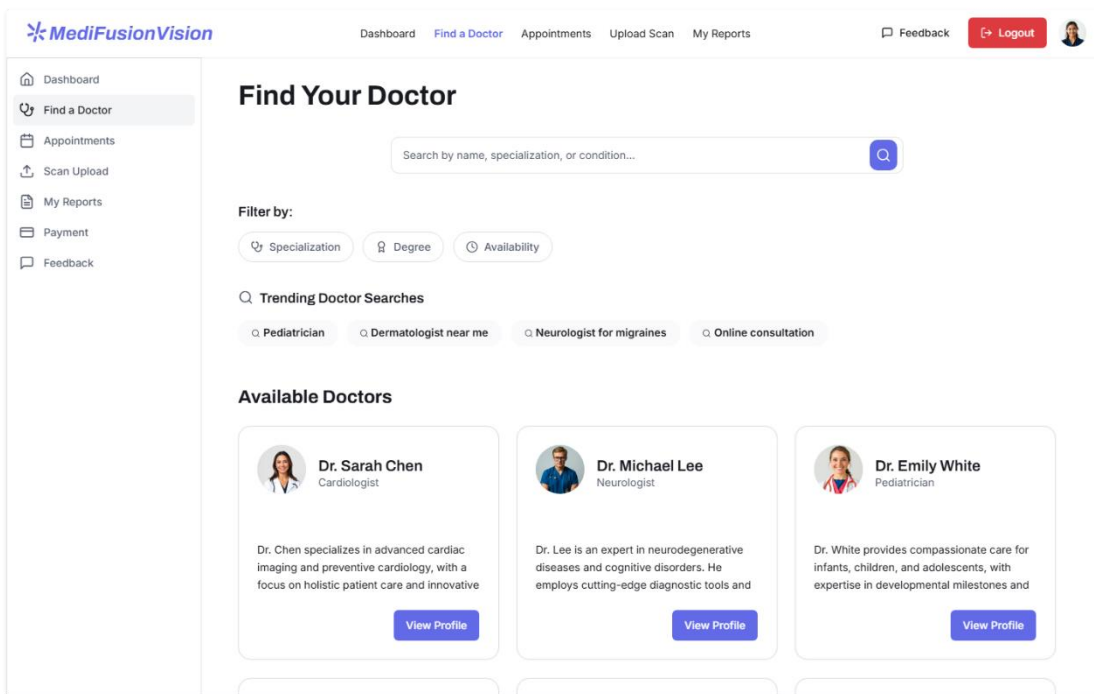


Figure 2-16: Doctor Search Screen

Table 2-14: Functional Requirements derived from Mockup M12

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Search bar	FR12.1 The system shall allow searching for doctors by name, specialization, or condition.	Only verified users may search doctors.

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Specialization, Degree, Availability filters	FR12.2 The system shall offer filters to refine doctor searches.	Only one filter category may be active at once.
Trending Doctor Searches	FR12.3 The system shall show popular or trending search terms for quick selection.	Suggestions based on recent search analytics.
Available Doctors list	FR12.4 The system shall display brief profiles for each available doctor.	Only currently available doctors accepting patients shown.
View Profile button	FR12.5 The system shall display full doctor profile details when selected.	Only patient users can view doctor profiles.

M13-Doctor Profile Screen

Patients view a detailed doctor profile (photo, specialty, reviews, background, qualifications), and can book an appointment.

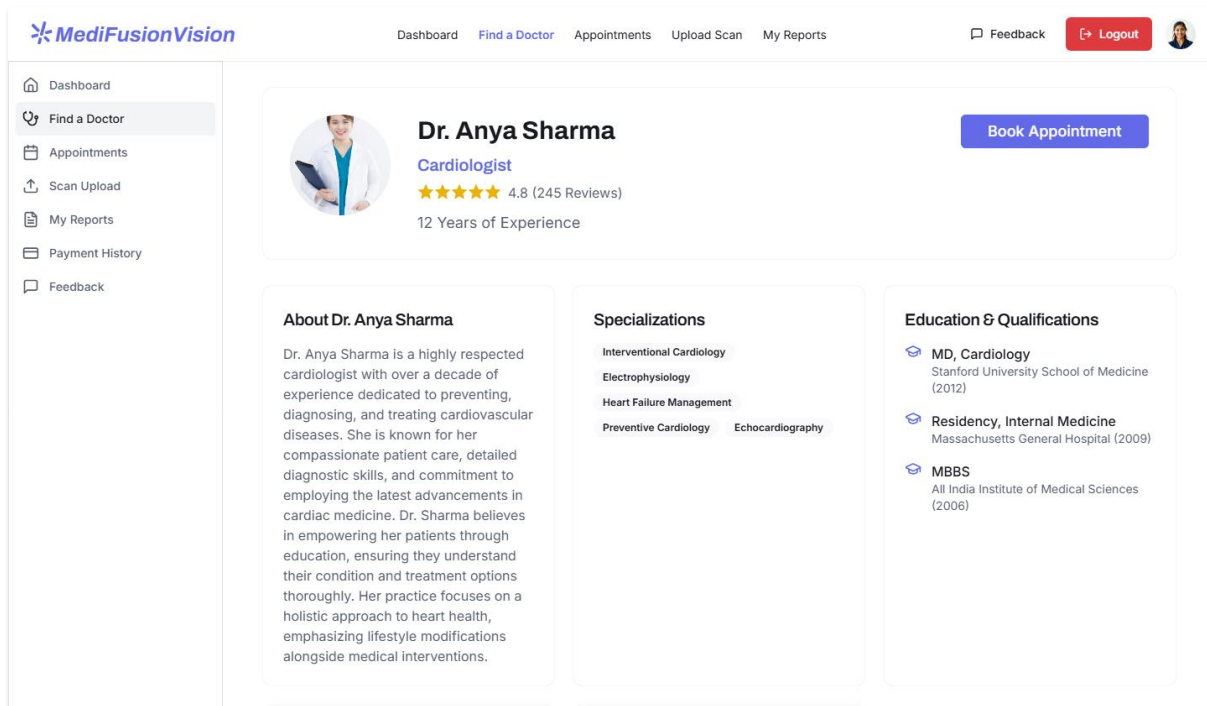


Figure 2-17: Doctor Profile Screen

Table 2-15: Functional Requirements derived from Mockup M13

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Doctor photo, name, specialty	FR13.1 The system shall display doctor's image, full name, and specialty.	Shown only for active doctors.
Ratings and reviews	FR13.2 The system shall show average rating and review count per doctor.	Reviews only by verified patients.
Experience, education, work history	FR13.3 The system shall show doctor's experience and education background.	Data editable only by doctor/admin.
Book Appointment button	FR13.4 The system shall allow initiating booking for a listed doctor.	Enabled only if doctor has available time slots.

M14-Appointment Booking Screen

Patients select their doctor, date, and consultation slot, record the reason, and confirm with payment details.

The screenshot displays the 'Book Your Appointment' screen in the MediFusionVision application. The top navigation bar includes links for Dashboard, Find a Doctor, Appointments (highlighted), Upload Scan, My Reports, Feedback, and a Logout button. The left sidebar lists navigation options: Dashboard, Find a Doctor, Appointments (selected), Scan Upload, My Reports, Payment, and Feedback. The main content area features a 'Book Your Appointment' section with a subtitle 'Select your preferred date, time, and payment method to confirm your consultation.' Below this, a doctor's profile for Dr. Anya Sharma, a Cardiologist, is shown. The 'Select Date' section displays 'October 13th, 2025'. The 'Select Time Slot' section shows a grid of time slots, with '10:00 AM - 10:30 AM' selected. The 'Reason for Consultation' section includes a text area containing the text 'Routine check-up and discussion about recent test results.'

Figure 2-18: Appointment Booking Screen

Table 2-16: Functional Requirements derived from Mockup M14

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Doctor card (name, specialty)	FR14.1 The system shall show selected doctor details for appointment.	Only active and available doctors are eligible.
Select Date field	FR14.2 The system shall let patients pick a valid consultation date.	Only future, available dates are selectable.
Select Time Slot buttons	FR14.3 The system shall present free time slots for appointments.	Only unbooked slots are shown.
Reason for Consultation field	FR14.4 The system shall require the reason for visit entry.	Field required for submission.
Payment Method option	FR14.5 The system shall collect and process payment method for appointments.	Payment required prior to confirmation.

M15-Payment Confirmation Screen

Patients receive payment summary, transaction info, method used, receipt and service details after booking.

The screenshot shows the 'Payment Confirmation' screen of the MediFusionVision application. The header includes the logo and navigation links: Dashboard, Find a Doctor, Appointments, Upload Scan, My Reports, Feedback, and a Logout button. A sidebar on the left lists the main menu items, with 'Payment' currently selected. The main content area is titled 'Payment Confirmation' and includes a thank-you message. It is divided into four sections: 'Transaction Summary' (showing Transaction ID, Date, and Service Type), 'Payment Method' (showing Card Type, Card Number, and Expiry Date), 'Service Details' (showing a doctor's photo, name, specialization, appointment date, time, and location), and 'Patient Information' (showing Patient Name, Patient ID, Email, and Phone). A 'Need Help?' section is also present at the bottom right.

Transaction Summary	
Transaction ID	TXN789012345
Date	October 26, 2024, 10:30 AM
Service Type	Doctor Consultation

Payment Method	
Card Type	Visa
Card Number	**** * 1234
Expiry Date	12/26

Service Details	
	
Doctor	Dr. Anya Sharma
Specialization	Cardiology
Appointment Date	October 28, 2024
Appointment Time	11:00 AM - 11:30 AM
Location	MediFusion Clinic, Room 305

Patient Information	
Patient Name	Jane Doe
Patient ID	PID98765
Email	jane.doe@example.com
Phone	+1 (555) 123-4567

Need Help?
If you have any questions regarding this payment or your appointment, please

Figure 2-19: Payment Confirmation Screen

Table 2-17: Functional Requirements derived from Mockup M15

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Transaction Summary section	FR15.1 The system shall show payment transaction summary and confirmation.	Receipt only after payment is completed.
Service Details section	FR15.2 The system shall display details of the purchased consultation.	Details only of the current payment.
Payment Method info	FR15.3 The system shall present card type, masked card number, and expiry date.	Sensitive information is masked for privacy.
Patient Info card	FR15.4 The system shall display patient's name, ID, email, and phone for verification.	Info visible only to logged-in patient.

M16-Scan Upload Screen

Patients upload supported medical scans (MRI, Retina, etc.), choose file, and trigger AI analysis for diagnostics.

MediFusionVision Dashboard Find a Doctor Appointments **Upload Scan** My Reports Feedback [Logout](#)

Dashboard Find a Doctor Appointments **Scan Upload** My Reports Payment Feedback

Upload Medical Scans

Select the type of medical scan you wish to upload, then drag and drop your files or browse your device. Our AI will analyze your scans to provide accurate diagnostic insights.

MRI Scan Retina Scan Text Report

Drag & drop your MRI files here, or

[Browse Files](#)

Supported formats: .dcm, .nii, .nrrd

[Upload Scan](#) [Run AI Analysis](#)

Quick Links Company Legal

Figure 2-20: Scan Upload Screen

Table 2-18: Functional Requirements derived from Mockup M16

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Scan type tabs (MRI/Retina/Report)	FR16.1 The system shall let patients select the scan type before uploading a file.	Only supported scan types can be chosen.
Drag & drop/Browse file area	FR16.2 The system shall accept scans dragged-and-dropped or selected for upload.	Formats must be .dcm, .nii, .nrrd only.
Upload Scan button	FR16.3 The system shall upload the chosen scan to the server.	Upload button enabled only for supported formats.
Run AI Analysis button	FR16.4 The system shall enable AI analysis after successful upload.	Only possible after upload is complete.

M17-AI Analysis Results Screen

Shows AI diagnostic results for uploaded scan: visualization, probability score, downloadable report, and sharing options.

MediReport

Dashboard
My Reports
Appointments
Medications
Messages
Profile

Final Diagnostic Report

Approved by: Dr. Eleanor Vance, MD
Approved on: October 26, 2023
Patient: Sophia Chen
Approved

Patient Demographics

Patient ID MRN-1234567	Date of Birth 1993-07-21	Gender Female
Contact (555) 123-4567	Email sophia.chen@example.com	

Clinical Summary
A concise overview of the key findings and patient status.

This report summarizes the recent comprehensive metabolic panel for Ms. Sophia Chen. The results indicate elevated fasting glucose and HbA1c levels, consistent with uncontrolled Type 2 Diabetes Mellitus. There are also signs of dyslipidemia with elevated total and LDL cholesterol, and low HDL cholesterol. Blood pressure readings remain within hypertensive range. Ms. Chen's overall health requires immediate attention to dietary and lifestyle modifications, alongside adherence to prescribed medications to mitigate long-term complications.

Figure 2-21: AI Analysis Results Screen

Table 2-19: Functional Requirements derived from Mockup M17

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
GradCAM Visualization panel	FR17.1 The system shall display AI GradCAM visualization for the scan analyzed.	Only for compatible scan types.
AI Prediction Summary	FR17.2 The system shall show diagnostic label, probability, and summary.	Displayed only after analysis is finished.
Download Full Report button	FR17.3 The system shall let patients download complete AI result reports.	Download enabled only after analysis completion.
Share with Doctor button	FR17.4 The system shall permit patients to instantly share results with their doctor.	Functional only if report is available.
Request Second Opinion button	FR17.5 The system shall let patients request a specialist review after result.	Button enabled only if initial result shown.
Scan Information card	FR17.6 The system shall display basic details of the scan just analyzed.	Shows just-uploaded scan information only.

M-18-Feedback Screen

Patients provide overall satisfaction rating, aspect feedback via sliders, and optionally share written comments.

MediFusionVision Dashboard Find a Doctor Appointments Upload Scan My Reports Feedback Logout

Dashboard Find a Doctor Appointments Scan Upload My Reports Payment Feedback Settings

Provide Your Feedback & Rate Your Experience
Your insights help us improve MediFusion Vision for everyone. Please share your thoughts and rate your overall satisfaction.

Overall Satisfaction
★★★★★

Ease of Navigation (80%)

Doctor Interaction (90%)

AI Report Clarity (70%)

Share Your Detailed Comments
Tell us about your experience with MediFusion Vision, what you liked, and where we can improve...

Submit Feedback

Quick Links Company Legal

Figure 2-22: Feedback Screen

Table 2-20: Functional Requirements derived from Mockup M18

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Overall Satisfaction stars	FR18.1 The system shall allow patients to rate their experience with star ratings.	Only completed consultations can be rated.
Navigation, Doctor, AI sliders	FR18.2 The system shall allow detailed scores for navigation, doctor, and AI clarity.	Sliders set before feedback submission.
Detailed Comments field	FR18.3 The system shall let patients enter freetext comments about their experience.	Optional; field can be left blank.
Submit Feedback button	FR18.4 The system shall store and submit patient's feedback in the system.	Only stores if mandatory fields are filled.

M19-Dashboard Overview (Admin)

The admin dashboard shows high-level stats (doctors, patients, appointments, reports) and visual analytics on clinic activity and scan distributions.

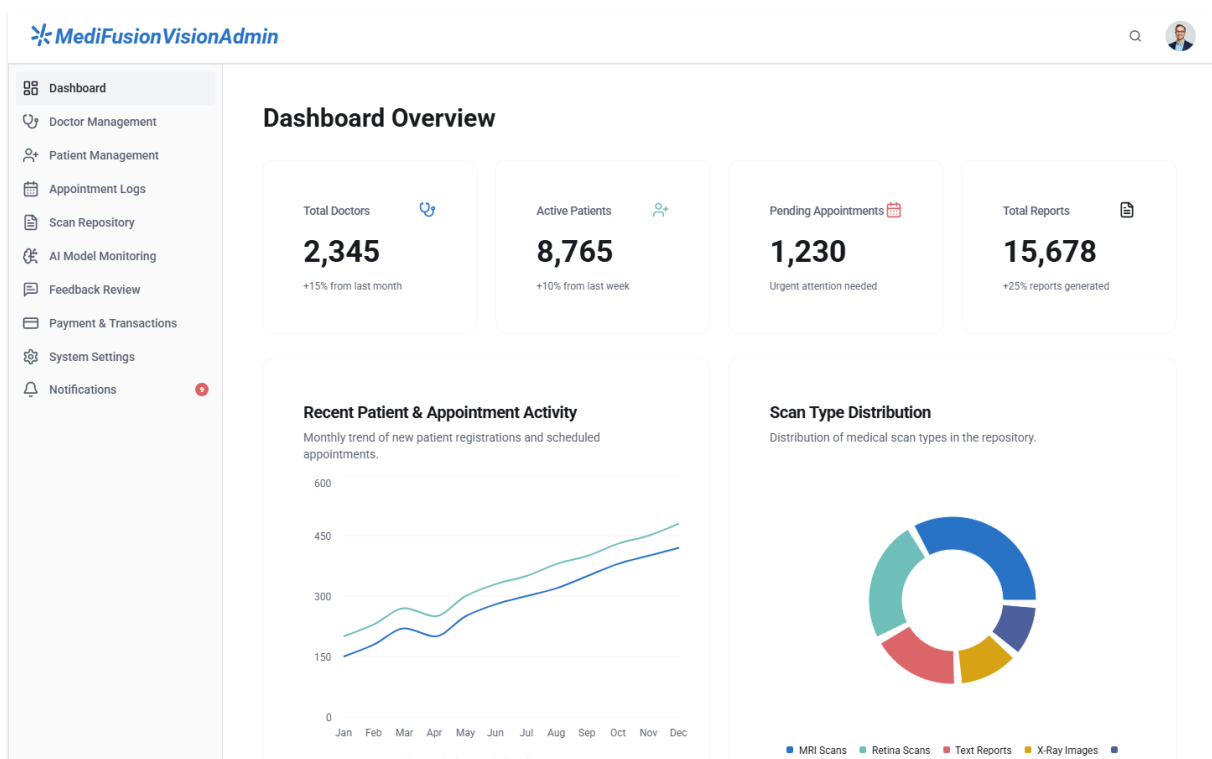


Figure 2-23: Dashboard Overview – Admin

Table 2-21: Functional Requirements derived from Mockup M19

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Total Doctors, Active Patients, etc. cards	FR19.1 The system shall display real-time counts for doctors, patients, appointments, and reports.	Counts pulled from backend databases.
Recent Patient & Appointment Activity chart	FR19.2 The system shall show time-series charts for patient registrations and appointments.	Chart data refreshed periodically.
Scan Type Distribution chart	FR19.3 The system shall visualize the breakdown of scan types stored on platform.	Visuals update in sync with current scan repository.

M20-Doctor Management Screen

Facilitates admin review, approval, editing, or rejection of doctors, with stats and searchable doctor cards.

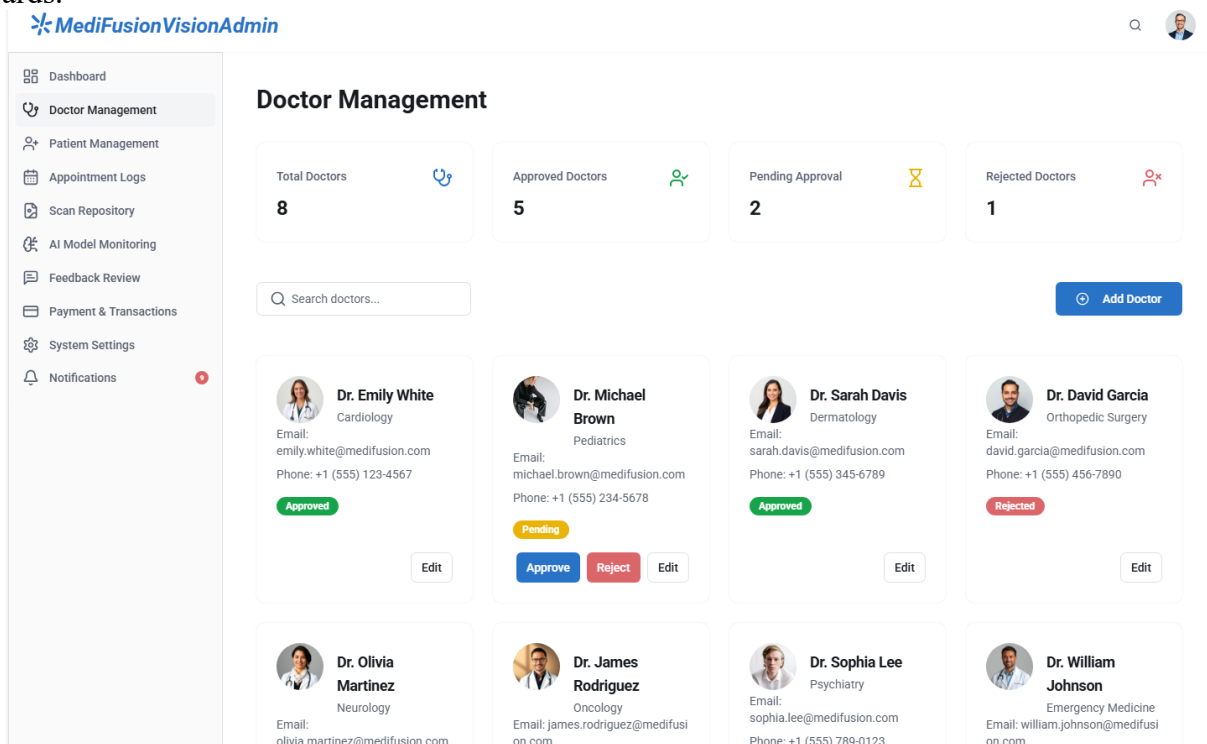


Figure 2-24: Doctor Management Screen

Table 2-22: Functional Requirements derived from Mockup M20

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Stats (Total, Approved, etc.)	FR20.1 The system shall show total, approved, pending, and rejected doctor counts.	Doctor status auto-calculates based on verification.
Doctor search bar	FR20.2 The system shall allow searching doctors by name or email.	Search returns matches on key fields only.
Doctor status badges (Approved/Pending/Rejected)	FR20.3 The system shall display status indicators per doctor card.	Badge updates as admin actions occur.
Approve / Reject buttons	FR20.4 The system shall enable admin to approve or reject a doctor's registration.	Only pending doctors can be approved/rejected.
Edit button	FR20.5 The system shall let admin edit doctor profile details.	Editable only for approved/pending doctors.
Add Doctor button	FR20.6 The system shall provide a button for registering a new doctor in the system.	Only admins can use this function.

M21-Patient Management Screen

Enables admins to search, list, update, and manage all patients, with filtering and status indicators.

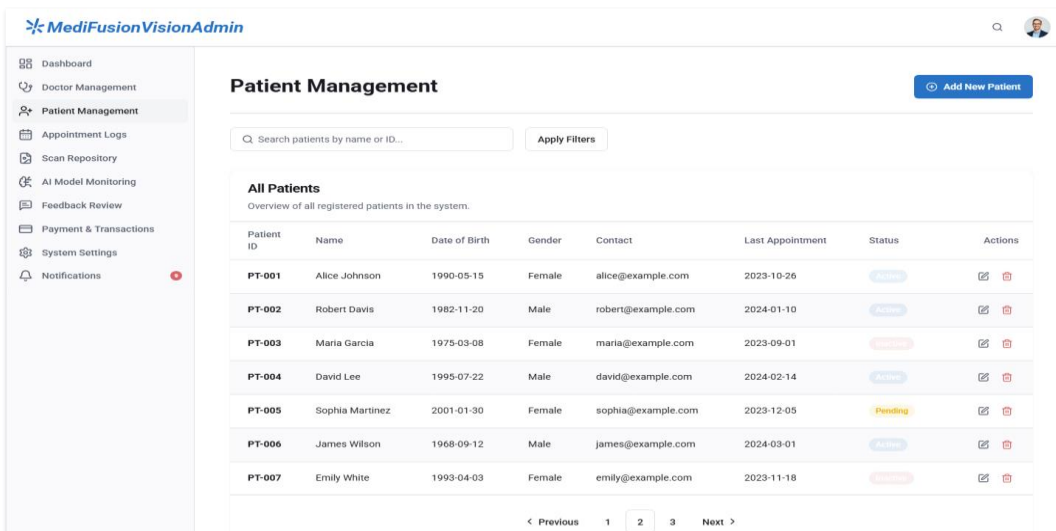


Figure 2-25: Patient Management Screen

Table 2-23: Functional Requirements derived from Mockup M21

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Search bar	FR21.1 The system shall enable searching of patients by name or ID.	Search limited to authorized admin only.
Apply Filters button	FR21.2 The system shall let admin filter patient list by various criteria (status, date).	Only one filter set can be active at a time.
All Patients table	FR21.3 The system shall display a paginated table of patient profiles and contact info.	List refreshes on new/edited data entry.
Patient status badges	FR21.4 The system shall display current status (Active, Pending, Inactive) for patients.	Status reflects record and appointment state.
Edit, Delete icons	FR21.5 The system shall let admin edit or remove a patient profile.	Deletion may require confirmation dialog.
Add New Patient button	FR21.6 The system shall provide an option to add a new patient manually.	Only admins can perform patient addition.
Pagination controls	FR21.7 The system shall paginate patient list as per system page size configuration.	Default page size set in settings.

M22-Appointment Logs

Grants admins access to all appointment history, with powerful filters and direct action viewing.

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Appointment status badges	FR22.5 The system shall indicate state of each appointment (Confirmed/Pending/Completed/etc).	Status pulled from backend for accuracy.
View Details action	FR22.6 The system shall provide action to see detailed view of a selected appointment.	Only for appointments in admin's permitted scope.

M23-Scan Repository Screen

Shows and manages all patient scan files in the system, with filters, views, and direct actions.

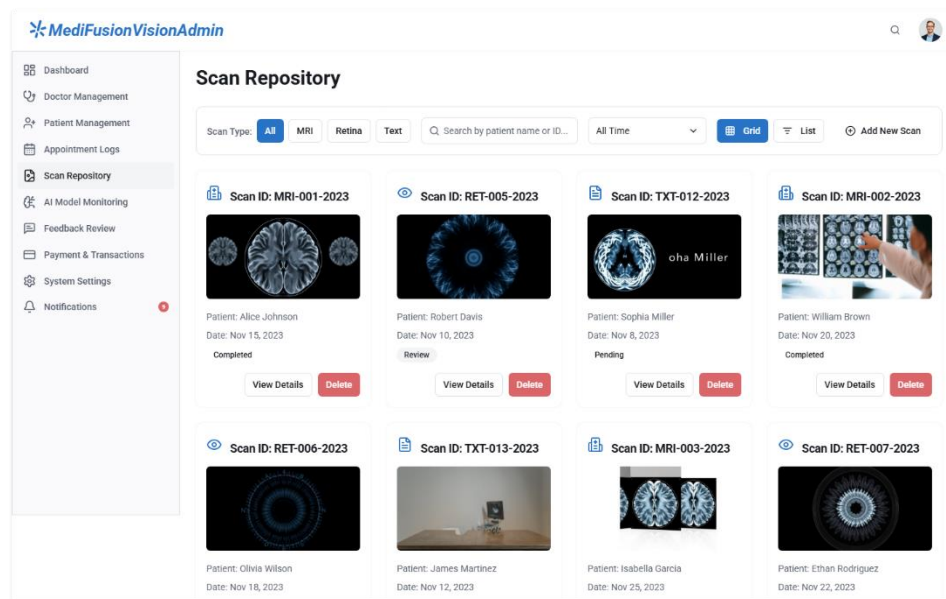


Figure 2-27: Scan Repository Screen

Table 2-25: Functional Requirements derived from Mockup M23

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Scan Type & Time filters	FR23.1 The system shall filter scan records by type and upload date.	Only supported types can filter data.
Search bar	FR23.2 The system shall enable searching by patient name or scan ID.	Search matches current filter selection.

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Grid/List view toggle	FR23.3 The system shall offer grid or list presentations of the scan records.	Preference may be saved per admin.
Scan cards (ID, Status, Patient, etc.)	FR23.4 The system shall display scan with visual details and metadata.	Scan ID and Patient reference mandatory on display.
View Details button	FR23.5 The system shall provide a details view for each scan.	Details may include full scan metadata.
Delete scan button	FR23.6 The system shall give the option to delete scan records after confirmation.	Only admins can delete, and deletion is irreversible.
Add New Scan button	FR23.7 The system shall provide a function to upload/ingest a new scan into repository.	Only supported formats allowed.

M24-AI Model Monitoring

Admins monitor real-time and historical AI model metrics (accuracy, latency, loss, F1 score), analyze trends, and investigate deployments.

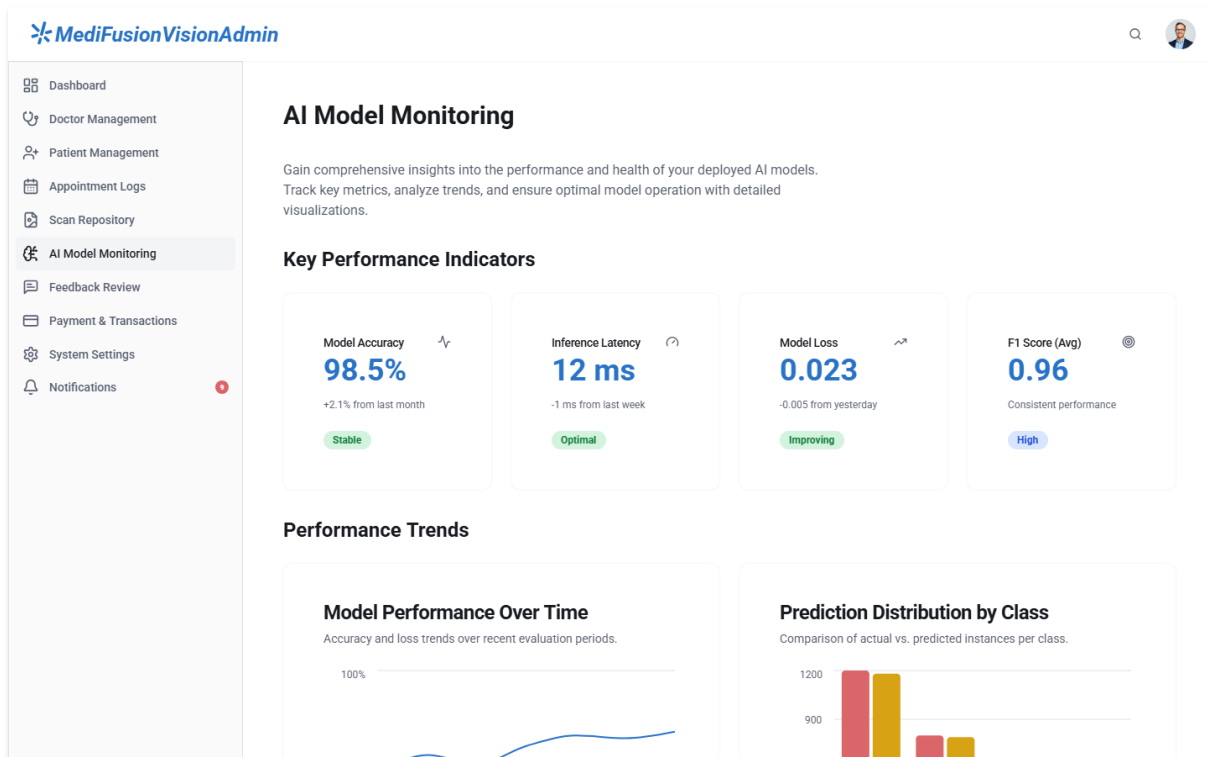


Figure 2-28: AI Model Monitoring Screen

Table 2-26: Functional Requirements derived from Mockup M24

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Model metrics cards (Accuracy, etc.)	FR24.1 The system shall show current model performance metrics (accuracy, latency, loss, F1 score).	Metrics update in real time from AI backend.
Performance Trends chart	FR24.2 The system shall visualize model performance trend over custom time windows.	Chart time windows selectable by admin.
Distribution charts	FR24.3 The system shall display prediction distribution by class for all recent scans.	Chart data includes only completed AI results.

M25-Notification Management

Notification center enables admins to review, search, and manage all system/user notifications with bulk actions and filtering.

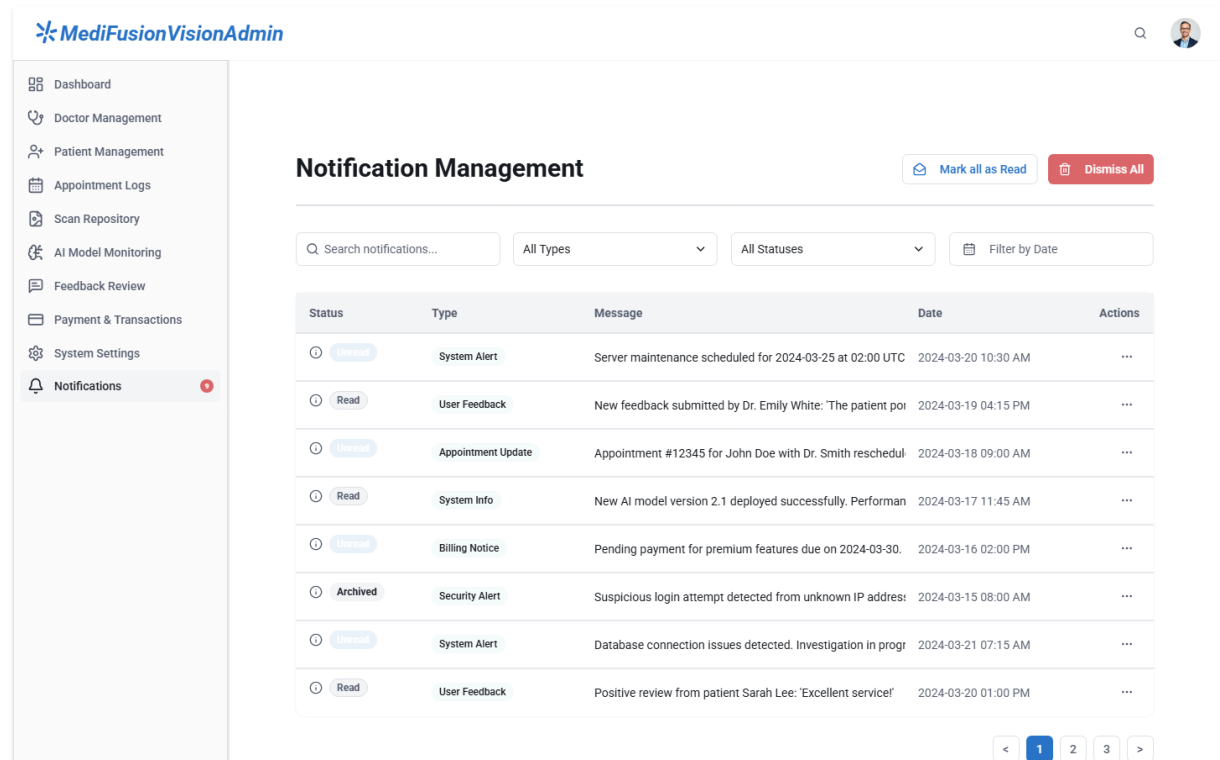


Figure 2-29: Notification Management Screen

Table 2-27: Functional Requirements derived from Mockup M25

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Search & filter controls	FR25.1 The system shall permit search and filter of notifications by text, type, or status.	Only records in current filter shown.
Notification list	FR25.2 The system shall display status, type, message, date, and actions for each notification.	Supports paging and real time updates.
Mark all as Read, Dismiss All	FR25.3 The system shall provide bulk actions to mark all as read or dismiss all notifications.	Actions affect only items visible to current admin.
Individual notification actions	FR25.4 The system shall let admin perform actions on specific notifications (read, archive, etc.).	Only appropriate actions available per status.

M26-Payment & Transaction History

Admins can track, filter, and export the payment and transaction records for all activities in the system.

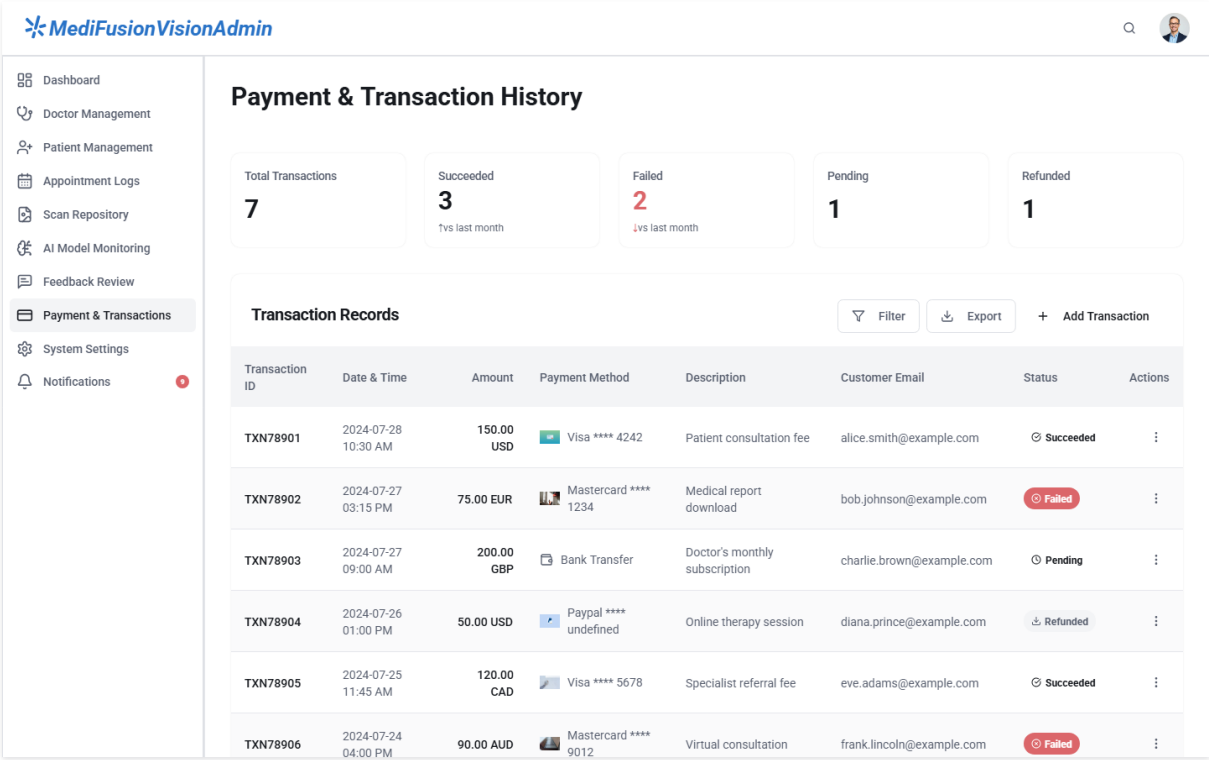


Figure 2-30: Payment & Transaction History Screen

Table 2-28: Functional Requirements derived from Mockup M26

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Transaction summary cards	FR26.1 The system shall show counts of total, succeeded, failed, pending, and refunded transactions.	Cards refresh as records are added or updated.
Filter, Export controls	FR26.2 The system shall enable filtering by date, status, and export of selected transactions.	Export only allowed for authorized admin roles.
Transaction table	FR26.3 The system shall list transaction details with ID, date/time, method, amount, description, customer, and status.	Table supports sorting and pagination.
Add Transaction button	FR26.4 The system shall allow adding a transaction manually, subject to proper validation.	Used for adjustments or error corrections.

M27-Feedback Review

Admins review, search, filter, and export user and staff feedback for response and reporting purposes.

ID	Date	User	Subject	Type	Status	Sentiment
FB001	2024-07-20	Dr. Elena Rodriguez	Issue with MRI scan upload feature	Bug Report	New	Negative
FB002	2024-07-19	Patient John Doe	Suggestion: Add online appointment booking	Feature Request	Reviewed	Positive
FB003	2024-07-18	Admin Assistant Sarah Lee	Query about patient data privacy settings	General Inquiry	New	Neutral
FB004	2024-07-17	Dr. Michael Chen	AI model misidentified a retina scan lesion	Bug Report	New	Negative
FB005	2024-07-16	Patient Jane Smith	Praise for intuitive dashboard design	General Inquiry	Archived	Positive
FB006	2024-07-15	Dr. Amelia G. Sharma	Request for multi-language support in reports	Feature Request	New	Positive
FB007	2024-07-14	Patient David Lee	Slow loading times on patient profile page	Bug Report	Reviewed	Negative

Figure 2-31: Feedback Review Screen

Table 2-28: Functional Requirements derived from Mockup M27

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Search & filter controls	FR27.1 The system shall support searching and filtering by status, type, or sentiment.	Multiple criteria can be combined for filtering.
Recent User Feedback table	FR27.2 The system shall display feedback details (ID, date, user, subject, status, sentiment).	Feedback updated as new records arrive.
Export Data button	FR27.3 The system shall permit exporting feedback data for records review.	Export is restricted to admin role users only.
View Feedback action	FR27.4 The system shall allow viewing the complete details of feedback entries.	Only admins/staff with permission may view all data.

M28-System Settings

Admins manage global settings, localization, user roles, security preferences, and the data retention policy for MediFusion Vision.

MediFusionVisionAdmin

System Settings

General | User Roles | Security | Notifications

Application Details
Configure basic application information and localization settings.

Application Name
MediFusion Vision Admin

Default Language
English

Time Zone
UTC-05:00 (Eastern Time)

Data Management & Retention
Define policies for data storage and automated backups.

Data Retention Policy
3 Years

Save General Settings

Figure 2-32: System Settings Screen

Table 2-30: Functional Requirements derived from Mockup M28

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Application Details section	FR28.1 The system shall show and allow editing of application name, language, and timezone.	Only admin-level users may modify these settings.
Tabs for User Roles, Security, etc.	FR28.2 The system shall categorize settings by general, roles, security, and notifications.	Only valid settings can be submitted.
Data Management & Retention section	FR28.3 The system shall display and let admin set retention policy for stored data.	Policy changes apply to new and existing records.
Save Settings button	FR28.4 The system shall save all changes upon confirmation by admin.	Changes take effect immediately unless otherwise specified.

M29-Splash Screen (Mobile)

Displays the MediFusion Vision logo and tagline during app startup.

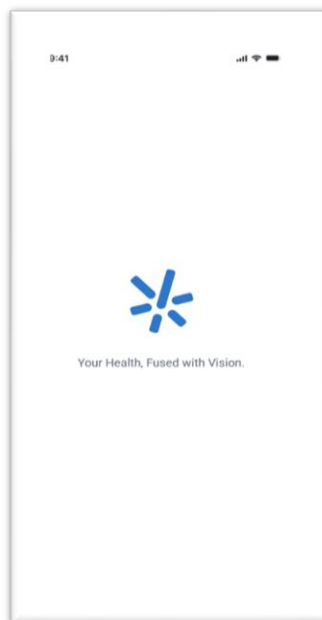


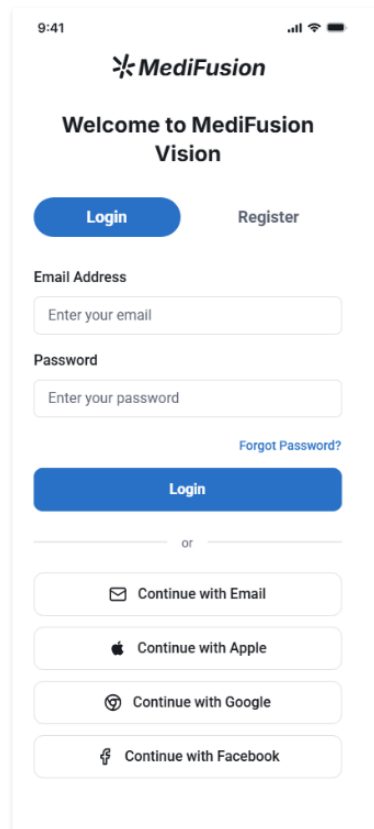
Figure2-33: Splash Screen – Mobile

Table 2-31: Functional Requirements derived from Mockup M29

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Splash logo and tagline	FR29.1 The system shall display the MediFusion Vision logo and tagline during app launch.	Shown until initial loading completes.

M30-Authentication Screen (Mobile)

Allows patients to log in or register using email/password or external providers; also supports password recovery.



9:41

MediFusion

Welcome to MediFusion Vision

Login Register

Email Address

Enter your email

Password

Enter your password

[Forgot Password?](#)

Login

or

Continue with Email

Continue with Apple

Continue with Google

Continue with Facebook

Figure 2-34: Authentication Screen – Mobile

Table 2-32: Functional Requirements derived from Mockup M30

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Email and Password fields	FR30.1 The system shall let users enter an email and password for authentication.	Both fields required for login.
Login button	FR30.2 The system shall authenticate users and grant access to the app upon valid credentials.	Only active, verified accounts can log in.
Forgot Password link	FR30.3 The system shall provide access to password reset workflow.	Security verification is required for reset.
Login/Register tabs	FR30.4 The system shall allow switching between login and registration forms.	Only one form active at a time.
Continue with Email/Apple/Google/Facebook	FR30.5 The system shall provide alternate social login options.	Only available for supported providers.

M31-Home Dashboard Screen (Mobile)

Welcomes the patient after login and provides quick access to appointments, AI reports, scan upload, and doctor messages.

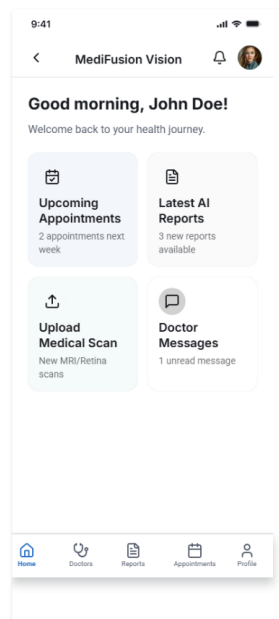


Figure 2-35: Home Dashboard Screen – Mobile

Table 2-33: Functional Requirements derived from Mockup M31

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Welcome message with user name	FR31.1 The system shall display a personalized greeting to the logged-in patient.	Visible only to authenticated users.
Upcoming Appointments shortcut	FR31.2 The system shall show a count of upcoming appointments and link to details.	Count updates in real-time based on scheduled visits.
Latest AI Reports shortcut	FR31.3 The system shall show recent AI diagnostic reports and link to full report history.	Only patient's own reports are displayed.
Upload Medical Scan shortcut	FR31.4 The system shall provide access to scan upload functionality.	Only enabled for verified user accounts.
Doctor Messages shortcut	FR31.5 The system shall indicate unread messages and link to chat with doctors.	New message count updates dynamically.
Bottom navigation bar	FR31.6 The system shall provide navigation to Home, Doctors, Reports, Appointments, Profile screens.	All pages accessible according to user session.

M32-Find a Doctor Screen (Mobile)

Patients can search for doctors, filter by specialization, and view top-rated doctors with profile and booking options.

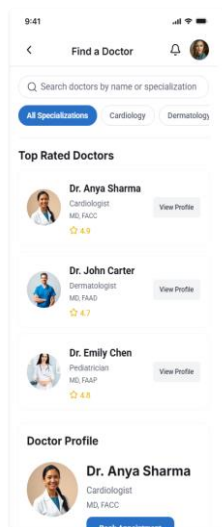


Figure 2-36: Find a Doctor Screen – Mobile

Table 2-34: Functional Requirements derived from Mockup M32

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Doctor search bar	FR32.1 The system shall let patients search for doctors by name or specialization.	Search results limited to active doctors only.
Specialization filter buttons	FR32.2 The system shall provide filters (e.g. Cardiology, Dermatology) to narrow doctor results.	Only one filter active at a time.
Top Rated Doctors cards	FR32.3 The system shall display top-rated doctor cards with name, specialty, rating, and profile.	Ratings based on verified patient feedback.
View Profile button	FR32.4 The system shall open the selected doctor's profile on tap.	Only for doctors currently accepting appointments.
Doctor Profile section & Book Appointment	FR32.5 The system shall show detailed doctor profiles and allow booking appointments if available.	Booking enabled only for available time slots.

M33-My Appointments Screen (Mobile)

Lists all upcoming and past appointments with doctor, specialty, status, as well as options to reschedule or cancel.

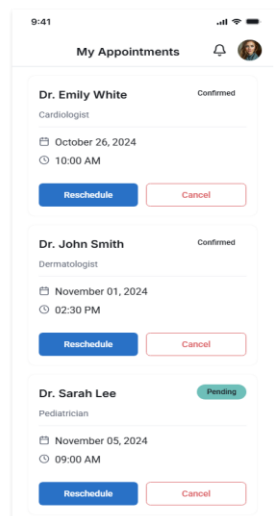


Figure 2-37: My Appointments Screen – Mobile

Table 2-35: Functional Requirements derived from Mockup M33

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Appointment cards (doctor, date, time, specialty)	FR33.1 The system shall display all appointments with doctor name, specialty, status, date, and time.	Only patient's own appointments shown.
Appointment status badges (Confirmed, Pending, Cancelled)	FR33.2 The system shall visually indicate appointment status.	Badge updates according to backend status.
Reschedule button	FR33.3 The system shall allow rescheduling an appointment to a new date and time.	Reschedule allowed only if status is Pending/Confirmed
Cancel button	FR33.4 The system shall allow the patient to cancel an upcoming appointment.	Cannot cancel if appointment is already completed.

M34-Book Appointment Screen (Mobile)

Handles selection of doctor, date, and available time slots for booking a new appointment, with confirmation.

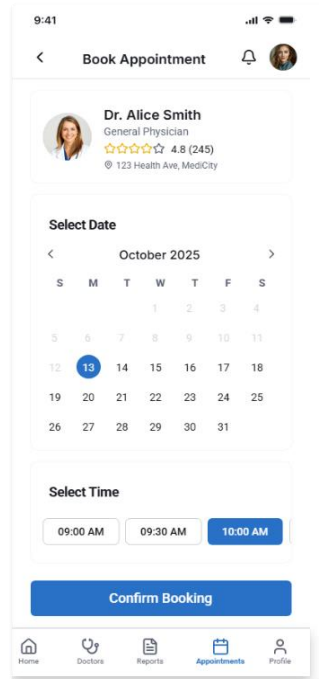


Figure 2-38: Book Appointment Screen – Mobile

Table 2-36: Functional Requirements derived from Mockup M34

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Doctor profile card	FR34.1 The system shall display selected doctor's name, specialty, location, and rating.	Only available doctors are shown.
Date selection calendar	FR34.2 The system shall provide a calendar to pick the desired appointment date.	Only future dates and available slots selectable.
Time slot buttons	FR34.3 The system shall list and allow selection of available time slots.	Only unbooked and valid slots enabled.
Confirm Booking button	FR34.4 The system shall finalize the booking after user confirms date and time.	Booking processed only if slot is still available.

M35-Upload Scan Screen (Mobile)

Enables the patient to select scan type, drag-and-drop or browse for files, and upload a medical scan for further analysis.

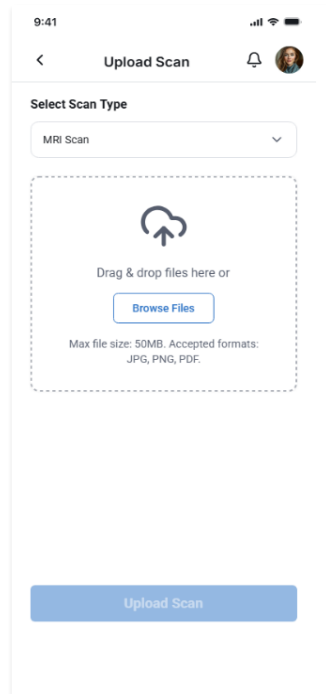


Figure 2-39: Upload Scan Screen – Mobile

Table 2-37: Functional Requirements derived from Mockup M35

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Select Scan Type dropdown	FR35.1 The system shall allow patients to select the scan type from a dropdown menu.	Only supported scan types appear.
Drag & drop / Browse Files area	FR35.2 The system shall support drag-and-drop and file selection from device storage.	Only files matching accepted types and size limit are allowed.
Max file size and format info	FR35.3 The system shall display file type and size restrictions to the patient.	Upload enabled only for compliant files.
Upload Scan button	FR35.4 The system shall upload the selected file to the server on user command.	Button is enabled only when a valid file is selected.

M36-AI Report Screen (Mobile)

Displays a specific AI-generated health report, with patient info, risk assessment, diagnostic findings, and status indicators for each section.

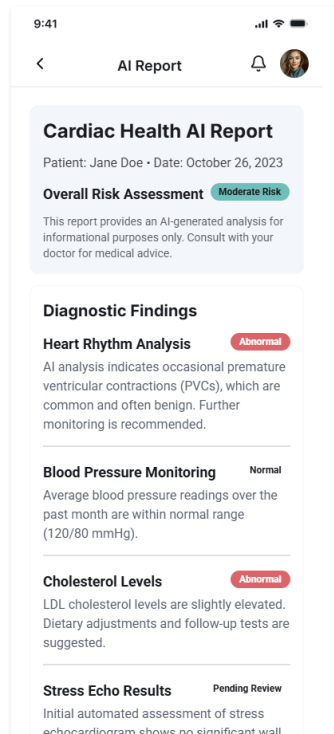


Figure 2-40: AI Report Screen – Mobile

Table 2-38: Functional Requirements derived from Mockup M36

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Report title and metadata	FR36.1 The system shall display the AI report title, patient name, and report date.	Only accessible to report owner and authorized roles.
Overall Risk Assessment with label	FR36.2 The system shall present the report's overall AI risk assessment with status badge.	Status label reflects AI analysis severity.
Diagnostic Findings section	FR36.3 The system shall list each diagnostic finding with explanation and result label.	Each finding status (Normal/Abnormal/Pending) as strong badge.
Status badges for findings	FR36.4 The system shall visually indicate normal, abnormal, or pending review for each diagnostic result.	Status is updated in real time on results.
Section for explanatory notes	FR36.5 The system shall display detailed explanations or recommendations for each finding.	Recommendations included as provided by system or doctor.

M37-Notifications & Profile Settings (Mobile)

Aggregates system notifications (appointments, reports, app updates) and allows toggling alert settings for different notification types.

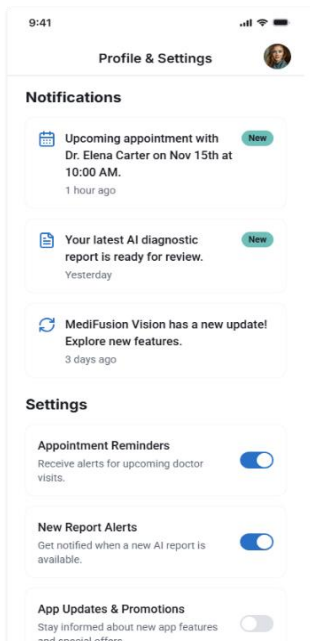


Figure 2-41: Notifications & Profile Settings – Mobile

Table 2-39: Functional Requirements derived from Mockup M37

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
System notifications list	FR37.1 The system shall display all user notifications by type, date, and status.	New notifications are highlighted visually.
Appointment and Report Alert toggles	FR37.2 The system shall allow patients to enable/disable notification types.	App applies changes immediately; persistent per user.
App Updates & Promotions toggle	FR37.3 The system shall allow opting in/out of feature and promotional notifications.	Changes reflect for future notifications only.

M38-Consultation Chat Screen (Mobile)

Provides a chat interface between patient and doctor for consultation and follow-up discussions, including support for text messages and timestamps.

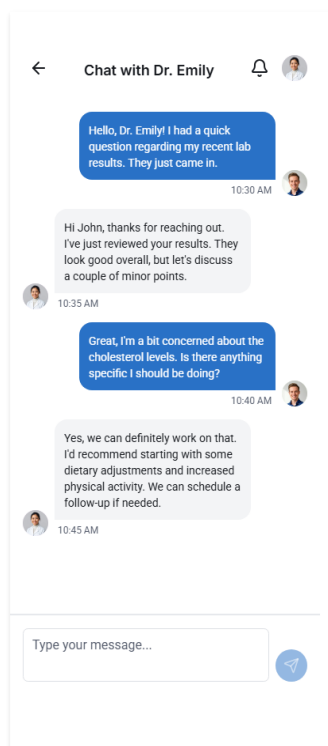


Figure 2-42: Consultation Chat Screen

Table 2-40: Functional Requirements derived from Mockup M38

Feature (from UI element)	Functional Requirement (FR-ID Statement)	Business Rule
Chat message bubbles (sent/received)	FR38.1 The system shall display all messages in a threaded conversation format.	Only chat for current consultation/session shown.
Message input field and send button	FR38.2 The system shall allow the patient to type and send messages instantly.	Send button enabled only with non-empty message.
Doctor and patient avatars	FR38.3 The system shall display avatar images for each participant in the chat.	Avatars reflect profile photos where set.
Timestamps on messages	FR38.4 The system shall show the time each message was sent or received.	Times shown in user's local timezone.

2.3 Non-Functional Requirements

This section outlines the non-functional requirements of the MediFusion Vision, covering key quality attributes such as reliability, usability, performance, security, availability, scalability, and maintainability. Each requirement is defined in clear, measurable, and verifiable terms to ensure effective testing and validation during system evaluation.

2.3.1 Reliability

REL-1: The system shall be deployed on a cloud-based, containerized infrastructure to ensure continuous operation and scalability across all hospital services..

REL-2: The system shall maintain an estimated Mean Time Between Failures (MTBF) of at least 300 hours under normal operating conditions.

REL-3: Any unexpected downtime or interruption in the backend, database, or AI diagnostic service shall be automatically detected and handled as a system failure event.

REL-4: In case of a server or database crash, all incomplete transactions shall be automatically rolled back to maintain data integrity.

REL-5: Once the system is restored, all previously committed records shall remain safe, consistent, and recoverable to ensure data durability.

REL-6: If the AI diagnostic model encounters an error, the system shall log the issue and display a user-friendly error message, allowing the doctor to retry or re-upload the scan.

REL-7: If the AI diagnostic model encounters an error, the system shall log the issue and display a user-friendly error message, allowing the doctor to retry or re-upload the scan.

REL-8: Automated monitoring and restart policies shall ensure that any affected service is recovered within 5 minutes of a detected failure, maintaining overall system reliability.

2.3.2 Usability

USE-1: New users (patients or doctors) shall be able to learn to use the core features (login, upload scan, view report) within **3 minutes** of guided onboarding.

USE-2: The user interface shall follow a consistent color scheme, iconography, and layout across both web and mobile platforms.

USE-3: The system shall support two languages (English and Urdu) for all patient-facing content.

USE-4: Accessibility options shall include text-to-speech, screen reader support, and high-contrast display modes for visually impaired users

USE-5: Error messages shall be displayed in plain, user-understandable language with suggestions for corrective actions.

2.3.3 Performance

PER-1: The AI diagnostic engine shall generate disease predictions for a single scan within 10 seconds on a standard GPU-enabled server.

PER-2: 95% of all pages and dashboard views shall load within 4 seconds on a 20 Mbps or faster internet connection.

PER-3: The system shall support at least 500 concurrent active users without performance degradation greater than 10%.

PER-4: System response time for database queries shall not exceed 2 seconds for standard patient record retrieval.

2.3.4 Security

SEC-1: All patient and doctor data shall be encrypted both in transit (HTTPS) and at rest (AES-256 encryption).

SEC-2: Each user shall have access only to their authorized modules using role-based access control (RBAC).

SEC-3: The system shall automatically log out users after 15 minutes of inactivity.

SEC-4: All login attempts, data uploads, and diagnosis events shall be recorded in a secure audit log.

SEC-5: The system shall require multi-factor authentication for doctors and admins.

2.4 External Interface Requirements

This section defines the external interfaces required for the MediFusion Vision system to interact effectively with users, AI services, and external APIs. These interfaces ensure smooth communication, secure data handling, and a consistent user experience across all platforms and devices..

2.4.1 User Interfaces Requirements

This section defines the logical characteristics of the user interfaces required by the MediFusion Vision system. Some possible items to include are

UI-1:The system will notify the User and Doctors before appointment

UI-2:The system shall display a pop-up box to tell the User when an error has occurred.

UI-3:The User shall be easily able to navigate through the system.

UI-4:The interface shall maintain consistency in fonts, color palette and icon styles across all screens.

UI-5:Standard navigation elements (Dashboard, Appointments, Reports, Logout) shall appear on every page.

UI-6:Screen layouts shall support a minimum resolution of 1366x768 (web) and responsive scaling for mobile devices

2.4.2 Software interfaces

The **MediFusion Vision** depends on several external software services and components to provide its core functionality, including database management, AI model integration, video consultation, payment processing, cloud deployment, and data visualization. These interfaces ensure seamless interaction between system modules and third-party APIs while maintaining performance, reliability, and data security. The following software interface requirements specify how the system interacts with these components:

SI-1: PostgreSQL v15 (Database Management System)

SI-1.1: The system shall use PostgreSQL v15 as the primary relational database to store patient, doctor, and administrative data securely.

SI-1.2: All medical records, diagnosis logs, and audit trails shall be stored using encrypted tablespaces to ensure HIPAA-compliant data protection.

SI-1.3: PostgreSQL has been selected because it offers strong relational integrity, advanced indexing, and compliance with healthcare data security standards, including HIPAA and Pakistan's PDPA 2023.

SI-1.4: Selected as the main database because it offers strong relational integrity, advanced indexing, and compliance with healthcare data security requirements

SI-2: PyTorch 2.x (AI Diagnostic Framework)

SI-2.1: The backend shall integrate with PyTorch models for disease detection from MRI, retinal, and X-ray images.

SI-3: Jitsi Meet API (Virtual Consultation Service)

SI-3.1: The system shall integrate Jitsi Meet API v2.0 to enable secure, real-time video consultations between patients and doctors.

*SI-3.2: The backend shall generate **unique meeting links** per session with role-based access (doctor/patient).*

SI-3.3: All video sessions shall be encrypted using Jitsi's E2E encryption and shall not store media data on the system.

SI-3.4: Used for virtual doctor-patient consultations because it is an open-source and HIPAA-compliant video conferencing API that ensures end-to-end encryption and privacy.

SI-4: Stripe API (Payment Gateway Service)

SI-4.1: The system shall integrate with the Stripe API to handle secure online payments for consultations and medical reports.

SI-4.2: The backend shall use Stripe's Payment Intents to create, confirm, and verify transactions through API calls.

SI-4.3: The frontend (React/Flutter) shall use Stripe Checkout to collect payment details without storing card data locally.

SI-4.4: All transactions shall use SSL encryption and follow PCI-DSS security standards

SI-4.5: Chosen for secure online payments and subscription management. It supports encrypted transactions and global payment standards suitable for medical platforms.

SI-5: Docker Engine & Docker Compose v26 (Deployment Environment)

SI-5.1: The system shall use Docker containers for modular deployment of backend, frontend, AI model service, and database.

SI-5.2: Docker Compose shall manage multi-container orchestration to ensure version control and consistent environment replication.

SI-5.3: Each service shall have a dedicated container image with isolated dependencies to prevent conflicts across environments.

SI-5.4: Used for deployment and environment isolation. It provides modular, scalable, and reproducible containers for each service (frontend, backend, AI model).

SI-6: Flask / Node.js Backend Framework

SI-6.1: The backend shall be implemented using **Flask (Python)** or **Node.js (Express)** for managing API routes, authentication, and AI model communication.

SI-6.2: The backend shall expose RESTful APIs for data exchange with frontend and mobile applications.

SI-6.3: The system shall use JWT-based token authentication for secure access control across all user roles.

SI-6.4: Preferred for backend API handling due to lightweight performance, integration ease, and compatibility with AI model services.

SI-7: React / Flutter Framework (Frontend Platform)

SI-7.1: The system shall employ React (web) and/or Flutter (mobile) frameworks to provide cross-platform accessibility.

SI-7.2: The frontend shall consume REST APIs exposed by the backend to display diagnostic reports, appointment data, and payment status.

SI-7.3: The interface shall support responsive layouts for desktop, tablet, and mobile devices.

SI-7.4: Used for building responsive and interactive user interfaces across devices while ensuring maintainability and component reuse.

SI-9: Cloud Storage Service (AWS S3 / Google Cloud Storage)

SI-9.1: The system shall store medical images, reports, and diagnostic outputs on secure cloud storage.

SI-9.2: The backend shall handle file upload/download using pre-signed URLs to prevent unauthorized access.

SI-9.3: All stored data shall be encrypted at rest using AES-256 encryption.

2.4.3 Hardware interfaces

This section describes the characteristics of each interface between the software and hardware components used by the MediFusion Vision. These interfaces define how the system interacts with device hardware to support image uploads, AI-based analysis, and real-time communication among patients, doctors, and administrators.

HI-1: *The system shall support patient devices (PCs, laptops, or smartphones) for registration, image upload, and report viewing.*

HI-2: *The system shall support doctor devices (computers or tablets) for accessing AI results and managing consultations.*

HI-3: *The system shall support admin computers for user management, data monitoring, and system control.*

HI-4: *The backend shall run on cloud or local servers with GPU support for AI model execution and data storage.*

2.4.4 Communications interfaces

The **MediFusion Vision** requires reliable and secure communication channels to ensure smooth interaction among patients, doctors, and administrators. Notifications, alerts, and updates shall be transmitted using in-app messaging, email, or SMS services to keep users informed about appointments, reports, and account activities. The following communication interface requirements define how the system will handle these communications:

CI-1: *The system shall send an appointment reminder to the patient one day before their scheduled consultation with the doctor.*

CI-2: *The system shall display an in-app message or notification to the patient once their diagnostic report becomes available for viewing.*

CI-3: *The system shall send a confirmation email to users upon successful registration to the system.*